

This is a working notes focusing on deriving the c -recursion relation for 2D $\mathcal{N} = 1$ super-conformal blocks in arbitrary plumbing channel. Written by AI (an individual), not by AI (artificial intelligence).

1 Virasoro conformal blocks

We first review the recipe for Virasoro conformal blocks, following [Cho-Collier-Yin], with the genus-2 conformal block in the theta channel as an example. The derivation can be generalized to arbitrary genus and arbitrary plumbing channel. In this case, the Virasoro conformal block is defined as

$$F(h_1, h_2, h_3, c) = \sum_{|A|=|B|, |C|=|D|, |E|=|F|} q_1^{|A|} q_2^{|C|} q_3^{|E|} G_{h_1}^{AB} G_{h_2}^{CD} G_{h_3}^{EF} \times \rho(L_{-A}\nu_1, L_{-C}\nu_2, L_{-E}\nu_3) \rho(L_{-B}\nu_1, L_{-D}\nu_2, L_{-F}\nu_3), \quad (1.1)$$

where A is a partition of $n = |A|$ in descending order, L_{-A} stands for a string of lowering operators made of L_{-m} 's, G_h^{AB} is the inverse of the Gram matrix $\langle h | L_{-A}^\dagger L_{-B} | h \rangle$, and ρ is the 3 point function of Virasoro descendents of the canonically normalized primaries ν_i with the structure constant stripped off:

$$\langle \xi_3 \otimes \bar{\xi}_3 | \xi_2 \otimes \bar{\xi}_2(z, \bar{z}) | \xi_1 \otimes \bar{\xi}_1 \rangle = C_{321} \rho(\xi_3, \xi_2, \xi_1 | z) \rho(\bar{\xi}_3, \bar{\xi}_2, \bar{\xi}_1 | \bar{z}). \quad (1.2)$$

We will also abbreviate $\rho(\xi_1, \xi_2, \xi_3 | 1)$ as $\rho(\xi_1, \xi_2, \xi_3)$.

The strategy of computing this conformal block is to treat it as a meromorphic function in c at fixed h_i . The poles on the complex c -plane can be located at the values where the Gram matrix develops a null state, and the whole conformal block can be determined by its large- c behavior and the residue at these poles. Namely,

$$F(h_1, h_2, h_3, c) = F(h_1, h_2, h_3, c \rightarrow \infty) + \sum_{i, r \geq 2, s \geq 1} \frac{W_{rs}^{(i)}(h_1, h_2, h_3, c)}{c - c_{rs}(h_i)}. \quad (1.3)$$

Note that we are considering the case where all the internal weights h_i are different, hence there are no double poles from two internal multiplets admits a null state simultaneously.

The poles at $c = c_{rs}(h_i)$ arises when the i -th intermediate multiplet develops a null state. The Kac formula then tells us

$$c_{rs}(h) = 1 + 6 (b_{rs}(h) + b_{rs}(h)^{-1})^2, \quad b_{rs}(h)^2 = \frac{rs - 1 + 2h + \sqrt{(r-s)^2 + 4(rs-1)h + 4h^2}}{1 - r^2} \quad (1.4)$$

where $r = 2, 3, 4, \dots$, $s = 1, 2, 3, \dots$. Conversely, let's denote the weight h that develops a null state at level rs at fixed c by $d_{rs}(c)$. At this value of c , the multiplet with lowest weight state

$\nu_{d_{rs}}$ admits a null state $\chi_{rs} = \sum_{|M|=rs} \chi_{rs}^M L_{-M} \nu_{d_{rs}}$. In this case, the residue $W_{rs}^{(i)}$ receives only contributions from the descendants of this null state. For example when $i = 1$, one have

$$\begin{aligned}
W_{rs}^{(1)} &= -\frac{\partial c_{rs}(h_1)}{\partial h_1} \lim_{h_1 \rightarrow d_{rs}} \frac{h_1 - d_{rs}}{\|\sum_{|M|=rs} \chi_{rs}^M L_{-M} \nu_h\|^2} \\
&\times \sum_{|A|=|B|, |C|=|D|, |E|=|F|} q_1^{|A|+rs} q_2^{|C|} q_3^{|E|} G_{h_1+rs}^{AB} G_{h_2}^{CD} G_{h_3}^{EF} \\
&\times \rho(L_{-A} \chi_{rs}, L_{-C} \nu_2, L_{-E} \nu_3) \rho(L_{-B} \chi_{rs}, L_{-D} \nu_2, L_{-F} \nu_3).
\end{aligned} \tag{1.5}$$

Using the factorization relation

$$\begin{aligned}
\rho(L_{-N} \chi_{rs}, L_{-M} \nu_2, L_{-P} \nu_3) &= \rho(L_{-N} \nu_{d_{rs}+rs}, L_{-M} \nu_2, L_{-P} \nu_3) \rho(\chi_{rs}, \nu_2, \nu_3) \\
\rho(L_{-N} \nu_1, L_{-M} \chi_{rs}, L_{-P} \nu_3) &= \rho(L_{-N} \nu_1, L_{-M} \nu_{d_{rs}+rs}, L_{-P} \nu_3) \rho(\nu_1, \chi_{rs}, \nu_3) \\
\rho(L_{-N} \nu_1, L_{-M} \nu_2, L_{-P} \chi_{rs}) &= \rho(L_{-N} \nu_1, L_{-M} \nu_2, L_{-P} \nu_{d_{rs}+rs}) \rho(\nu_1, \nu_2, \chi_{rs}) \\
\rho(L_{-N} \chi_{rs}, L_{-M} \nu_2, L_{-P} \chi_{rs}) &= \rho(L_{-N} \nu_{d_{rs}+rs}, L_{-M} \nu_2, L_{-P} \nu_{d_{rs}+rs}) \rho(\chi_{rs}, \nu_2, \chi_{rs}),
\end{aligned} \tag{1.6}$$

one can then arrive at

$$W_{rs}^{(1)} = -\frac{\partial c_{rs}(h_1)}{\partial h_1} A_{rs}^c q_1^{rs} [P_c^{rs}(h_3, h_2)]^2 F(h_1 \rightarrow h_1 + rs, c \rightarrow c_{rs}(h_1)), \tag{1.7}$$

where the quantity A_{rs}^c and the fusion polynomial $P_c^{rs}(h_3, h_2)$ are defined as

$$\begin{aligned}
A_{rs}^c &= \lim_{h \rightarrow d_{rs}} \frac{h - d_{rs}}{\|\sum_{|M|=rs} \chi_{rs}^M L_{-M} \nu_h\|^2} = \frac{1}{2} \prod_{m=1-r}^r \prod_{n=1-s}^s (mb_{rs} + nb_{rs}^{-1})^{-1}, \quad (m, n) \neq (0, 0), (r, s) \\
P_c^{rs}(h_1, h_2) &\equiv \rho(\chi_{rs}, \nu_2, \nu_1) = \prod_{p=1-r}^{r-1} \prod_{q=1-s}^{s-1} \frac{\lambda_1 + \lambda_2 + pb_{rs} + qb_{rs}^{-1}}{2} \frac{\lambda_1 - \lambda_2 + pb_{rs} + qb_{rs}^{-1}}{2},
\end{aligned} \tag{1.8}$$

where the products for the fusion polynomial are taken over $p + r = 1(\text{mod } 2)$ and $q + s = 1(\text{mod } 2)$, and λ_i are defined by $h_i = \frac{1}{4}(b_{rs} + b_{rs}^{-1})^2 - \frac{1}{4}\lambda_i^2$. The other residues $W_{rs}^{(2)}, W_{rs}^{(3)}$ follows similarly.

The large- c limit $F(h_1, h_2, h_3, c \rightarrow \infty)$ of the conformal block can be computed by the following. Denoting by $\langle A \rangle$ the number of non- L_{-1} generators in L_{-A} . The key observation is that, the Gram matrix element G_{AB} scales like $c^{\langle A \rangle}$ for $A = B$, no faster than $c^{\langle A \rangle - 1}$ for $\langle A \rangle = \langle B \rangle$ but $A \neq B$, and no faster than $c^{\min(\langle A \rangle, \langle B \rangle)}$ for $\langle A \rangle \neq \langle B \rangle$. This can be directly seen from the Virasoro algebra, where the leading term in c is given by the commutators $[L_m, L_{-m}]$. One can then start from the basis of states $\{L_{-A} \nu, |A| = n\}$ at level n , apply Gram-Schmidt orthogonalization, and obtain a basis of orthogonal states $\mathcal{L}_{-A} \nu$. These states are of the form

$$\mathcal{L}_{-A} \nu = L_{-A} \nu + \sum_{|B|=n, \langle B \rangle \leq \langle A \rangle, B \neq A} g_{AB}(c, h) L_{-B} \nu, \tag{1.9}$$

where $g_{AB}(c, h)$ is order $\mathcal{O}(c^0)$ for $B < A$ and $\mathcal{O}(c^{-1})$ for $\langle B \rangle = \langle A \rangle$. In this basis, the conformal block can be written as

$$F(h_1, h_2, h_3, c) = \sum_{A,B,C} q_1^{|A|} q_2^{|B|} q_3^{|C|} \frac{\rho(\mathcal{L}_{-A}\nu_1, \mathcal{L}_{-B}\nu_2, \mathcal{L}_{-C}\nu_3)^2}{\|\mathcal{L}_{-A}\nu_1\|^2 \|\mathcal{L}_{-B}\nu_2\|^2 \|\mathcal{L}_{-C}\nu_3\|^2} \quad (1.10)$$

where $\|\mathcal{L}_{-A}\nu_1\|^2$ is of order $c^{\langle A \rangle}$.

Now let's consider the scaling behavior of $\rho(\mathcal{L}_{-A}\nu_1, \mathcal{L}_{-B}\nu_2, \mathcal{L}_{-C}\nu_3)$ in the $c \rightarrow \infty$ limit. In this limit, only the term of order $c^{(\langle A \rangle + \langle B \rangle + \langle C \rangle)/2}$ survives in the conformal block. For a general 3-point function $\rho(L_{-A}\nu_1, L_{-B}\nu_2, L_{-C}\nu_3)$, the leading term must come from pairwise contractions $[L_m, L_{-m}]$ of non- L_{-1} Virasoro generators, and the maximum number of such contractions is precisely $(\langle A \rangle + \langle B \rangle + \langle C \rangle)/2$, which gives a scaling behavior of $c^{(\langle A \rangle + \langle B \rangle + \langle C \rangle)/2}$. In each term of $\mathcal{L}_{-A}\nu$, only the $L_{-A}\nu$ term can give this scaling, since $g_{AB}(c, h)$ is subleading for $\langle B \rangle = \langle A \rangle$ and the correlator is subleading for $\langle B \rangle < \langle A \rangle$. Moreover, since only the central term in the commutator contributes to this limit, we have

$$\rho(\mathcal{L}_{-A}\nu_1, \mathcal{L}_{-B}\nu_2, \mathcal{L}_{-C}\nu_3) \rightarrow \rho(L_{-A'}\mathbf{1}, L_{-B'}\mathbf{1}, L_{-C'}\mathbf{1})\rho(L_{-1}^{k_A}\nu_1, L_{-1}^{k_B}\nu_2, L_{-1}^{k_C}\nu_3), \quad (1.11)$$

where $\mathbf{1}$ is the identity operator, and we have written $A = A' \cup \{1\}^{k_A}$. Similarly, the 2-point function also admits a similar factorization relation at large c , namely

$$\|\mathcal{L}_{-A}\nu\|^2 \rightarrow \|L_{-A'}\mathbf{1}\|^2 \|L_{-1}^{k_A}\nu\|^2. \quad (1.12)$$

Plugging these relations back to the conformal block, one can then arrive at the large- c limit

$$F(h_1, h_2, h_3, c \rightarrow \infty) = F(0, 0, 0, c \rightarrow \infty) \times F_{\mathfrak{sl}(2)}(h_1, h_2, h_3), \quad (1.13)$$

where the first factor is the large- c limit of the vacuum block, and the second factor is the global $\mathfrak{sl}(2)$ block that's independent of c . The global $\mathfrak{sl}(2)$ blocks are generally simple and each term in the q -sum can be computed in terms of Pochhammer symbols (See (4.19)-(4.22) of [Choi-Collier-Yin]), while the vacuum block in the Schottky frame has a simple product formula (See (5.5)).

Assembling all the pieces, we arrive at the final form for the c -recursion relation for genus-2 Virasoro conformal blocks in the theta channel:

$$\begin{aligned} F(h_1, h_2, h_3, c) &= F(0, 0, 0, c \rightarrow \infty) \times F_{\mathfrak{sl}(2)}(h_1, h_2, h_3) \\ &+ \sum_{r \geq 2, s \geq 1} - \left(\frac{\partial c_{rs}(h_1)}{\partial h_1} \right) \frac{A_{rs}^c q_1^{rs} [P_c^{rs}(h_3, h_2)]^2}{c - c_{rs}(h_1)} F(h_1 \rightarrow h_1 + rs, c \rightarrow c_{rs}(h_1)) \\ &+ 2 \text{ permutation in } h_1, h_2, h_3. \end{aligned} \quad (1.14)$$

Using this relation, one can recursively evaluate the conformal block $F(h_1, h_2, h_3, c)$ as a q -expansion, as the polar terms are all higher-order in q due to the factor q^{rs} .

2 $\mathcal{N} = 1$ superconformal algebra

Let $\nu \in \{0, \frac{1}{2}\}$. The two-dimensional $\mathcal{N} = 1$ super-Virasoro algebra SCA_ν is generated by

$$\{L_n, G_r \mid n \in \mathbb{Z}, r \in \mathbb{Z} + \nu\}.$$

Its nonvanishing (anti)commutation relations are

$$\begin{aligned} [L_m, L_n] &= (m - n)L_{m+n} + \frac{c}{12} (m^3 - m) \delta_{m+n,0}, \\ [L_m, G_r] &= \left(\frac{m}{2} - r\right) G_{m+r}, \\ \{G_r, G_s\} &= 2L_{r+s} + \frac{c}{3} \left(r^2 - \frac{1}{4}\right) \delta_{r+s,0}. \end{aligned} \tag{2.1}$$

Here, $\nu = 0$ and $\nu = \frac{1}{2}$ correspond to the Ramond and Neveu–Schwarz sectors, respectively.

A two-dimensional $\mathcal{N} = (1, 1)$ superconformal field theory contains both holomorphic and anti-holomorphic superconformal algebras. We denote the anti-holomorphic algebra by $\widetilde{\text{SCA}}_\nu$, with generators

$$\{\tilde{L}_n, \tilde{G}_r \mid n \in \mathbb{Z}, r \in \mathbb{Z} + \tilde{\nu}\}.$$

They obey

$$\begin{aligned} [\tilde{L}_m, \tilde{L}_n] &= (m - n)\tilde{L}_{m+n} + \frac{\tilde{c}}{12} (m^3 - m) \delta_{m+n,0}, \\ [\tilde{L}_m, \tilde{G}_r] &= \left(\frac{m}{2} - r\right) \tilde{G}_{m+r}, \\ \{\tilde{G}_r, \tilde{G}_s\} &= 2\tilde{L}_{r+s} + \frac{\tilde{c}}{3} \left(r^2 - \frac{1}{4}\right) \delta_{r+s,0}. \end{aligned} \tag{2.2}$$

The holomorphic and anti-holomorphic generators mutually commute.

3 NS superconformal blocks

We now generalize this c -recursion relation to superconformal blocks in $\mathcal{N} = 1$ superconformal field theories. The sphere 4-point case of the recursion relation has been worked out in [0611266, 0611295], and the generalization to general plumbing channel has been worked out in [1806.09563].

Apart from the bosonic moduli, one also needs to specify a spin structure to put a SCFT on a Riemann surface. A plumbing channel is specified by a choice of NS and R for each punctures, a modulus q and a sign η for each pair of plumbed punctures. Only NS and NS, R and R punctures are allowed to be plumbed together, and the NS/R configuration of punctured

on each 3-punctured sphere must be (NS, NS, NS) or (NS, R, R). For genus- g surface, there are 2^g choices of NS/R punctures and 2^g independent sign choices for each tube.

In this section, we only consider the case where all puncture are NS, where the c -recursion relation is more or less parallel to the Virasoro case. The NS sector supervirasoro algebra is given by

Similar to the notation $L_{-A}\nu$, we will denote a NS superconformal primary by ϕ , and its descendent by

$$\mathbf{L}_{-A}\phi \equiv L_{-n_1} \cdots L_{-n_\ell} G_{-r_1} \cdots G_{-r_p} \phi, \quad |A| \equiv \sum_{i=1}^{\ell} n_i + \sum_{j=1}^p r_j, \quad (-1)^A \equiv (-1)^p \quad (3.1)$$

where $n_1 \geq \dots \geq n_\ell, r_1 > \dots > r_p$. We will also take ϕ to be Grassmann even.

Yutai: I do not know why we can choose the parity of phi in general. I would assume it to be generic.

In the language of 2D $\mathcal{N} = 1$ superspace, one can pack ϕ and its level- $\frac{1}{2}$ descendent $\chi \equiv G_{-1/2}\phi$ in the primary superfield $\Phi(z, \theta)$, defined by

$$\Phi(z, \theta) = \phi(z) + \theta \chi(z). \quad (3.2)$$

Superconformal symmetry then constraints the 3-point functions to be of the following form

$$\begin{aligned} \langle \Phi_1(z_1, \theta_1) \Phi_2(z_2, \theta_2) \rangle &\propto \frac{\delta_{h_1 h_2}}{Z_{12}^{2h_1}}, \\ \langle \Phi_1(z_1, \theta_1) \Phi_2(z_2, \theta_2) \Phi_3(z_3, \theta_3) \rangle &= \frac{\langle \Phi_1(0, 0) \Phi_2(1, \Theta) \Phi_3'(\infty, 0) \rangle}{Z_{21}^{h_1+h_2-h_3} Z_{31}^{h_1+h_3-h_2} Z_{32}^{h_2+h_3-h_1}}, \end{aligned} \quad (3.3)$$

where

$$\begin{aligned} Z_{ij} &\equiv z_i - z_j - \theta_i \theta_j \\ \Theta &\equiv \frac{\theta_1 z_{23} + \theta_2 z_{31} + \theta_3 z_{12} - \frac{1}{2} \theta_1 \theta_2 \theta_3}{\sqrt{z_{12} z_{13} z_{23}}}. \end{aligned} \quad (3.4)$$

Importantly, there are now two independent structure constants for each triple of NS supermultiplets, they are the 1 and Θ component of the 3-point function, given essentially by $C_{123} = \langle \phi_1(0) \phi_2(1) \phi_3(\infty) \rangle$ and $\tilde{C}_{123} = \langle \phi_1(0) \chi_2(1) \phi_3(\infty) \rangle$. This is due to the fact that the 3-NS punctured genus-0 super-Riemann surface has 1 odd moduli, which is given precisely by this odd cross-ratio Θ .

For 3-point functions of superconformal descendants $\langle \mathbf{L}_{-A} \phi_1 \mathbf{L}_{-B} \phi_2 \mathbf{L}_{-C} \phi_3 \rangle$, It can be shown by superconformal Ward identity that the $|A| + |B| + |C| \in \mathbb{Z}$ case only depends on C_{123} , while the $|A| + |B| + |C| \in \mathbb{Z} + \frac{1}{2}$ case only depends on $\tilde{C}_{123}$. We can then define two types of

normalized 3-point functions

$$\langle\langle \xi_3 \otimes \bar{\xi}_3 | \xi_2 \otimes \bar{\xi}_2(z, \bar{z}) | \xi_1 \otimes \bar{\xi}_1 \rangle\rangle = (\pm) \begin{cases} C_{321} \rho_0(\xi_3, \xi_2, \xi_1 | z) \rho_0(\bar{\xi}_3, \bar{\xi}_2, \bar{\xi}_1 | \bar{z}), & \text{integer total level,} \\ \tilde{C}_{321} \rho_1(\xi_3, \xi_2, \xi_1 | z) \rho_1(\bar{\xi}_3, \bar{\xi}_2, \bar{\xi}_1 | \bar{z}), & \text{half-integer total level,} \end{cases} \quad (3.5)$$

by essentially normalizing

$$\rho_0(\phi_1, \phi_2, \phi_3 | 1) = 1, \quad \rho_1(\phi_1, \chi_2, \phi_3 | 1) = 1. \quad (3.6)$$

The sign $(\pm)$ is here indicating that we use the operator order $\langle \xi_3 \xi_2 \xi_1 \bar{\xi}_3 \bar{\xi}_2 \bar{\xi}_1 \rangle$ to define the ρ on the RHS, which can differ from the operator order in LHS by a sign. Again, we denote $\rho_a(\xi_1, \xi_2, \xi_3) = \rho_a(\xi_1, \xi_2, \xi_3 | 1)$.

The anti-holomorphic sector is entirely analogous to the holomorphic sector. We distinguish quantities in the anti-holomorphic sector by adding a tilde over them.

Conformal Block Decomposition of Genus 2 Partition Function

For illustration, we consider a 2d $\mathcal{N} = (1, 1)$ superconformal field theory and derive the conformal-block decomposition of its genus-two partition function. The final result present a nice holomorphic factorization structure after keeping track of the signs in the computation.

Let $\mathcal{O}_i(z, \bar{z})$ be the superconformal primary operator with conformal weight $(h_i, \tilde{h}_i)$. In the following derivation, for convenience, we will assume $\mathcal{O}_i$ to be orthogonal to each other, with

$$\langle\langle \mathcal{O}_i | \mathcal{O}_j \rangle\rangle = D_i \delta_{ij}. \quad (3.7)$$

Here $\langle\langle \psi |$ denotes the BPZ conjugate of state ψ .

Primary operator $\mathcal{O}_i(z, \bar{z})$ admits a formal factorization

$$\mathcal{O}_i(z, \bar{z}) = \sqrt{(-1)^{p_i \tilde{p}_i}} D_i \phi_i(z) \otimes \tilde{\phi}_i(\bar{z}), \quad (3.8)$$

with ϕ the holomorphic superconformal primary with parity p_i and $\tilde{\phi}(\bar{z})$ the anti-holomorphic superconformal primary with parity $\tilde{p}_i$, normalized such that

$$\langle\langle \phi | \phi \rangle\rangle = \langle\langle \tilde{\phi} | \tilde{\phi} \rangle\rangle = 1. \quad (3.9)$$

The parity of $\mathcal{O}_i$ is $p_i + \tilde{p}_i$.

Consider the genus-two partition function $Z_{g=2}$. We work in the theta-graph plumbing channel, taking all three tubes to have (NS, NS) spin structure in both the holomorphic and

anti-holomorphic sectors. In the theta plumbing channel, the genus-two partition function $Z_{g=2}$ has a prescription

$$\begin{aligned}
Z_{g=2}^\Theta = & \sum_{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3} \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F| \\ |\tilde{A}|=|\tilde{B}|, |\tilde{C}|=|\tilde{D}|, |\tilde{E}|=|\tilde{F}|}} \prod_{i=1}^3 q_i^{h_i} \tilde{q}_i^{\tilde{h}_i} \times q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \tilde{q}_1^{|\tilde{A}|} \tilde{\eta}_1^{\tilde{A}} \tilde{q}_2^{|\tilde{C}|} \tilde{\eta}_2^{\tilde{C}} \tilde{q}_3^{|\tilde{E}|} \tilde{\eta}_3^{\tilde{E}} \\
& \times (-1)^{(A+\tilde{A}+p_1+\tilde{p}_1)(C+\tilde{C}+p_2+\tilde{p}_2)+(A+\tilde{A}+p_1+\tilde{p}_1)(E+\tilde{E}+p_3+\tilde{p}_3)+(C+\tilde{C}+p_2+\tilde{p}_2)(E+\tilde{E}+p_3+\tilde{p}_3)} \\
& \times (\mathbb{G}_1)^{A\tilde{A};B\tilde{B}} (\mathbb{G}_2)^{C\tilde{C};D\tilde{D}} (\mathbb{G}_3)^{E\tilde{E};F\tilde{F}} \\
& \times \left\langle \left\langle \mathbf{L}_{-A} \tilde{\mathbf{L}}_{-\tilde{A}} \mathcal{O}_1 \middle| \mathbf{L}_{-C} \tilde{\mathbf{L}}_{-\tilde{C}} \mathcal{O}_2(1,1) \middle| \mathbf{L}_{-E} \tilde{\mathbf{L}}_{-\tilde{E}} \mathcal{O}_3 \right\rangle \right\rangle \times \left\langle \left\langle \mathbf{L}_{-B} \tilde{\mathbf{L}}_{-\tilde{B}} \mathcal{O}_1 \middle| \mathbf{L}_{-D} \tilde{\mathbf{L}}_{-\tilde{D}} \mathcal{O}_2(1,1) \middle| \mathbf{L}_{-F} \tilde{\mathbf{L}}_{-\tilde{F}} \mathcal{O}_3 \right\rangle \right\rangle
\end{aligned} \tag{3.10}$$

In this prescription, we need to choose the Weyl frame to be the one defined by theta channel plumbing. The sign in the sum comes from reordering the operators into the prescription of the matrix element. The gram matrix is defined by

$$(\mathbb{G}_i)_{A\tilde{A};B\tilde{B}} \equiv \left\langle \left\langle \mathbf{L}_{-A} \tilde{\mathbf{L}}_{-\tilde{A}} \mathcal{O}_i \middle| \mathbf{L}_{-B} \tilde{\mathbf{L}}_{-\tilde{B}} \mathcal{O}_i \right\rangle \right\rangle \tag{3.11}$$

It can be factorized into the holomorphic and anti-holomorphic pieces as

$$(\mathbb{G}_i)_{A\tilde{A};B\tilde{B}} = D_i (-1)^{(\tilde{A}+\tilde{p}_i)(B+p_i)+p_i\tilde{p}_i} (\mathbf{G}_{h_i})_{AB} (\tilde{\mathbf{G}}_{h_i})_{\tilde{A}\tilde{B}}. \tag{3.12}$$

where

$$(\mathbf{G}_{h_i})_{AB} := \left\langle \left\langle \mathbf{L}_{-A} \phi_i \middle| \mathbf{L}_{-B} \phi_i \right\rangle \right\rangle, (\tilde{\mathbf{G}}_{h_i})_{\tilde{A}\tilde{B}} := \left\langle \left\langle \tilde{\mathbf{L}}_{-\tilde{A}} \tilde{\phi}_i \middle| \tilde{\mathbf{L}}_{-\tilde{B}} \tilde{\phi}_i \right\rangle \right\rangle. \tag{3.13}$$

The three point function would also factorize

$$\begin{aligned}
& \left\langle \left\langle \mathbf{L}_{-A} \tilde{\mathbf{L}}_{-\tilde{A}} \mathcal{O}_1 \middle| \mathbf{L}_{-C} \tilde{\mathbf{L}}_{-\tilde{C}} \mathcal{O}_2(1,1) \middle| \mathbf{L}_{-E} \tilde{\mathbf{L}}_{-\tilde{E}} \mathcal{O}_3 \right\rangle \right\rangle \\
& \times \left\langle \left\langle \mathbf{L}_{-B} \tilde{\mathbf{L}}_{-\tilde{B}} \mathcal{O}_1 \middle| \mathbf{L}_{-D} \tilde{\mathbf{L}}_{-\tilde{D}} \mathcal{O}_2(1,1) \middle| \mathbf{L}_{-F} \tilde{\mathbf{L}}_{-\tilde{F}} \mathcal{O}_3 \right\rangle \right\rangle \\
& = (-1)^{p_1\tilde{p}_1+p_2\tilde{p}_2+p_3\tilde{p}_3} D_1 D_2 D_3 \left(C_{123}^{(a)} \right)^2 \\
& \times \rho_a(\mathbf{L}_{-A} \phi_1, \mathbf{L}_{-C} \phi_2, \mathbf{L}_{-E} \phi_3) \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{A}} \tilde{\phi}_1, \tilde{\mathbf{L}}_{-\tilde{C}} \tilde{\phi}_2, \tilde{\mathbf{L}}_{-\tilde{E}} \tilde{\phi}_3) \\
& \times \rho_a(\mathbf{L}_{-B} \phi_1, \mathbf{L}_{-D} \phi_2, \mathbf{L}_{-F} \phi_3) \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{B}} \tilde{\phi}_1, \tilde{\mathbf{L}}_{-\tilde{D}} \tilde{\phi}_2, \tilde{\mathbf{L}}_{-\tilde{F}} \tilde{\phi}_3).
\end{aligned} \tag{3.14}$$

Here $A + C + E + p_1 + p_2 + p_3 = \tilde{A} + \tilde{C} + \tilde{E} + \tilde{p}_1 + \tilde{p}_2 + \tilde{p}_3 = a \bmod 2$. In this step, the sign coming from the reordering of holomorphic primaries and anti-holomorphic primaries will cancel out because the same factor appears in both matrix element.

Using the factorialization properies, the genus 2 partition function simplifies into

$$\begin{aligned}
Z_{g=2} = & \sum_{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3} \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F| \\ |\tilde{A}|=|\tilde{B}|, |\tilde{C}|=|\tilde{D}|, |\tilde{E}|=|\tilde{F}|}} \prod_{i=1}^3 q_i^{h_i} \tilde{q}_i^{\tilde{h}_i} \times q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \tilde{q}_1^{|\tilde{A}|} \tilde{\eta}_1^{\tilde{A}} \tilde{q}_2^{|\tilde{C}|} \tilde{\eta}_2^{\tilde{C}} \tilde{q}_3^{|\tilde{E}|} \tilde{\eta}_3^{\tilde{E}} \\
& \times (-1)^{a+K+\bar{K}} \left(C_{123}^{(a)} \right)^2 \mathbf{G}_{h_1}^{AB} \mathbf{G}_{h_2}^{CD} \mathbf{G}_{h_3}^{EF} \tilde{\mathbf{G}}_{\tilde{h}_1}^{\tilde{A}\tilde{B}} \tilde{\mathbf{G}}_{\tilde{h}_2}^{\tilde{C}\tilde{D}} \tilde{\mathbf{G}}_{\tilde{h}_3}^{\tilde{E}\tilde{F}} \\
& \times \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3) \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{A}}\tilde{\phi}_1, \tilde{\mathbf{L}}_{-\tilde{C}}\tilde{\phi}_2, \tilde{\mathbf{L}}_{-\tilde{E}}\tilde{\phi}_3) \\
& \times \rho_a(\mathbf{L}_{-B}\phi_1, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3) \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{B}}\tilde{\phi}_1, \tilde{\mathbf{L}}_{-\tilde{D}}\tilde{\phi}_2, \tilde{\mathbf{L}}_{-\tilde{F}}\tilde{\phi}_3).
\end{aligned} \tag{3.15}$$

Here

$$\begin{aligned}
K &\equiv (A+p_1)(C+p_2) + (A+p_1)(E+p_3) + (C+p_2)(E+p_3) \pmod{2}, \\
\bar{K} &\equiv (\tilde{A}+\tilde{p}_1)(\tilde{C}+\tilde{p}_2) + (\tilde{A}+\tilde{p}_1)(\tilde{E}+\tilde{p}_3) + (\tilde{C}+\tilde{p}_2)(\tilde{E}+\tilde{p}_3) \pmod{2}.
\end{aligned} \tag{3.16}$$

Define the NS superconformal block by

$$\begin{aligned}
\mathbf{F}_a^\Theta(h_1, h_2, h_3, c) = & \sum_{|A|=|B|, |C|=|D|, |E|=|F|, (-1)^{A+C+E}=(-1)^a} q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \mathbf{G}_{h_1}^{AB} \mathbf{G}_{h_2}^{CD} \mathbf{G}_{h_3}^{EF} \\
& \times (-1)^K \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3) \rho_a(\mathbf{L}_{-B}\phi_1, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3),
\end{aligned} \tag{3.17}$$

and analogously the anti-holomorphic block $\tilde{F}_a$. The genus 2 partition function then reduces to

$$Z_{g=2} = \sum_{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3} \sum_{a=0,1} \left(C_{123}^{(a)} \right)^2 (-1)^a \prod_{i=1}^3 q_i^{h_i} \tilde{q}_i^{\tilde{h}_i} F_a(h_1, h_2, h_3, c) \tilde{F}_a(\tilde{h}_1, \tilde{h}_2, \tilde{h}_3, \tilde{c}). \tag{3.18}$$

We can also compute the genus 2 partition function in the glass channel. It is given by the following prescription

$$\begin{aligned}
Z_{g=2}^{Glass} = & \sum_{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3} \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F| \\ |\tilde{A}|=|\tilde{B}|, |\tilde{C}|=|\tilde{D}|, |\tilde{E}|=|\tilde{F}|}} \prod_{i=1}^3 q_i^{h_i} \tilde{q}_i^{\tilde{h}_i} \times q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \tilde{q}_1^{|\tilde{A}|} \tilde{\eta}_1^{\tilde{A}} \tilde{q}_2^{|\tilde{C}|} \tilde{\eta}_2^{\tilde{C}} \tilde{q}_3^{|\tilde{E}|} \tilde{\eta}_3^{\tilde{E}} \\
& \times (-1)^{(E+\tilde{E}+p_3+\tilde{p}_3)[(A+\tilde{A}+p_1+\tilde{p}_1)+(C+\tilde{C}+p_2+\tilde{p}_2)]} \\
& \times (\mathbb{G}_1)^{A\tilde{A}; B\tilde{B}} (\mathbb{G}_2)^{C\tilde{C}; D\tilde{D}} (\mathbb{G}_3)^{E\tilde{E}; F\tilde{F}} \\
& \times \left\langle \left\langle \mathbf{L}_{-A} \tilde{\mathbf{L}}_{-\tilde{A}} \mathcal{O}_1 \middle| \mathbf{L}_{-E} \tilde{\mathbf{L}}_{-\tilde{E}} \mathcal{O}_3(1, 1) \middle| \mathbf{L}_{-B} \tilde{\mathbf{L}}_{-\tilde{B}} \mathcal{O}_1 \right\rangle \times \left\langle \left\langle \mathbf{L}_{-C} \tilde{\mathbf{L}}_{-\tilde{C}} \mathcal{O}_2 \middle| \mathbf{L}_{-F} \tilde{\mathbf{L}}_{-\tilde{F}} \mathcal{O}_3(1, 1) \middle| \mathbf{L}_{-D} \tilde{\mathbf{L}}_{-\tilde{D}} \mathcal{O}_2 \right\rangle \right\rangle
\end{aligned} \tag{3.19}$$

The gram matrix factorize in the same way as (3.12). The three point function would factorize as

$$\begin{aligned}
& \left\langle \left\langle \mathbf{L}_{-A} \tilde{\mathbf{L}}_{-\tilde{A}} \mathcal{O}_1 \middle| \mathbf{L}_{-E} \tilde{\mathbf{L}}_{-\tilde{E}} \mathcal{O}_3(1,1) \middle| \mathbf{L}_{-B} \tilde{\mathbf{L}}_{-\tilde{B}} \mathcal{O}_1 \right\rangle \right. \\
& \quad \times \left. \left\langle \left\langle \mathbf{L}_{-C} \tilde{\mathbf{L}}_{-\tilde{C}} \mathcal{O}_2 \middle| \mathbf{L}_{-F} \tilde{\mathbf{L}}_{-\tilde{F}} \mathcal{O}_3(1,1) \middle| \mathbf{L}_{-D} \tilde{\mathbf{L}}_{-\tilde{D}} \mathcal{O}_2 \right\rangle \right. \\
& = (-1)^{p_1 \tilde{p}_1 + p_2 \tilde{p}_2 + p_3 \tilde{p}_3 + (A+p_1+a)(\tilde{A}+\tilde{p}_1+a) + (C+p_2+a)(\tilde{C}+\tilde{p}_2+a)} D_1 D_2 D_3 \left(C_{131}^{(a)} C_{232}^{(a)} \right) \\
& \quad \times \rho_a(\mathbf{L}_{-A} \phi_1, \mathbf{L}_{-E} \phi_3, \mathbf{L}_{-B} \phi_1) \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{A}} \tilde{\phi}_1, \tilde{\mathbf{L}}_{-\tilde{E}} \tilde{\phi}_3, \tilde{\mathbf{L}}_{-\tilde{B}} \tilde{\phi}_1) \\
& \quad \times \rho_a(\mathbf{L}_{-C} \phi_2, \mathbf{L}_{-F} \phi_3, \mathbf{L}_{-D} \phi_2) \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{C}} \tilde{\phi}_2, \tilde{\mathbf{L}}_{-\tilde{F}} \tilde{\phi}_3, \tilde{\mathbf{L}}_{-\tilde{D}} \tilde{\phi}_2).
\end{aligned} \tag{3.20}$$

where $a \equiv E + p_3 \equiv \tilde{E} + \tilde{p}_3 \pmod{2}$.

Using the factorialization properties, the glass channel genus 2 partition function can be expressed into

$$\begin{aligned}
Z_{g=2}^{\text{Gl}} &= \sum_{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3} \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F| \\ |\tilde{A}|=|\tilde{B}|, |\tilde{C}|=|\tilde{D}|, |\tilde{E}|=|\tilde{F}|}} \prod_{i=1}^3 q_i^{h_i} \tilde{q}_i^{\tilde{h}_i} q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E q_1^{\tilde{A}} \tilde{\eta}_1^{\tilde{A}} q_2^{\tilde{C}} \tilde{\eta}_2^{\tilde{C}} q_3^{\tilde{E}} \tilde{\eta}_3^{\tilde{E}} \\
& \quad \times C_{131}^{(a)} C_{232}^{(a)} (-1)^{a+K_{\text{Gl}}+\tilde{K}_{\text{Gl}}} \mathbf{G}_{h_1}^{AB} \mathbf{G}_{h_2}^{CD} \mathbf{G}_{h_3}^{EF} \tilde{\mathbf{G}}_{\tilde{h}_1}^{\tilde{A}\tilde{B}} \tilde{\mathbf{G}}_{\tilde{h}_2}^{\tilde{C}\tilde{D}} \tilde{\mathbf{G}}_{\tilde{h}_3}^{\tilde{E}\tilde{F}} \\
& \quad \times \rho_a(\mathbf{L}_{-A} \phi_1, \mathbf{L}_{-E} \phi_3, \mathbf{L}_{-B} \phi_1) \rho_a(\mathbf{L}_{-C} \phi_2, \mathbf{L}_{-F} \phi_3, \mathbf{L}_{-D} \phi_2) \\
& \quad \times \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{A}} \tilde{\phi}_1, \tilde{\mathbf{L}}_{-\tilde{E}} \tilde{\phi}_3, \tilde{\mathbf{L}}_{-\tilde{B}} \tilde{\phi}_1) \tilde{\rho}_a(\tilde{\mathbf{L}}_{-\tilde{C}} \tilde{\phi}_2, \tilde{\mathbf{L}}_{-\tilde{F}} \tilde{\phi}_3, \tilde{\mathbf{L}}_{-\tilde{D}} \tilde{\phi}_2).
\end{aligned}$$

where

$$K_{\text{Gl}} = a(A + p_1 + C + p_2), \quad \tilde{K}_{\text{Gl}} = a(\tilde{A} + \tilde{p}_1 + \tilde{C} + \tilde{p}_2). \tag{3.21}$$

Define the NS superconformal block in glass channel as

$$\begin{aligned}
\mathbf{F}_a^{\text{Gl}}(h_1, h_2, h_3, c) &= \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F|, (-1)^{E+p_3}=(-1)^a}} q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \mathbf{G}_{h_1}^{AB} \mathbf{G}_{h_2}^{CD} \mathbf{G}_{h_3}^{EF} \\
& \quad \times (-1)^{K_{\text{Gl}}} \rho_a(\mathbf{L}_{-A} \phi_1, \mathbf{L}_{-E} \phi_3, \mathbf{L}_{-B} \phi_1) \rho_a(\mathbf{L}_{-C} \phi_2, \mathbf{L}_{-F} \phi_3, \mathbf{L}_{-D} \phi_2).
\end{aligned} \tag{3.22}$$

and analogously the anti-holomorphic block $\tilde{F}_a^{\text{Gl}}$. The genus 2 partition function then reduces to

$$Z_{g=2} = \sum_{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3} \sum_{a=0,1} \left(C_{131}^{(a)} C_{232}^{(a)} \right) (-1)^a \prod_{i=1}^3 q_i^{h_i} \tilde{q}_i^{\tilde{h}_i} F_a^{\text{Gl}}(h_1, h_2, h_3, c) \tilde{F}_a^{\text{Gl}}(\tilde{h}_1, \tilde{h}_2, \tilde{h}_3, \tilde{c}). \tag{3.23}$$

c-Recursion for NS Block

We will keep using the theta channel as an example, with the plumbing data denoted by (q_i, η_i) . For simplicity, we will assume the holomorphic primary operators are grassmann even. The

all-NS conformal block is given by

$$\begin{aligned} \mathbf{F}_a(h_1, h_2, h_3, c) = & \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F|, (-1)^{A+C+E}=(-1)^a \\ \times (-1)^{AC+AE+CE}}} q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \mathbf{G}_{h_1}^{AB} \mathbf{G}_{h_2}^{CD} \mathbf{G}_{h_3}^{EF} \\ & \rho_a(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3) \rho_a(\mathbf{L}_{-B}\phi_1, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3), \end{aligned} \quad (3.24)$$

where $\mathbf{G}$ denotes the inverse of the super-Gram matrix $\langle h | \mathbf{L}_{-A}^\dagger \mathbf{L}_{-B} | h \rangle$. a takes value from 0, 1. The sign $(-1)^{AC+AE+CE}$ comes from moving the operator orders. The condition $(-1)^{A+C+E} = (-1)^a$ ensures that only total-integer level or total-half-integer level intermediate states enters the 3-point function.

The NS supermultiplet admits a level- $rs/2$ null state when

$$h_{rs}(c) = \frac{Q^2 - (rb + sb^{-1})^2}{8}, \quad Q = b + b^{-1} \quad (r \geq 2, s \geq 1, r + s \in 2\mathbb{Z}), \quad (3.25)$$

where b is defined by $c = \frac{3}{2} + 3(b + b^{-1})^2$. Conversely, for fixed h , the degeneration happens when the central charge is

$$c_{r,s}(h) = \frac{15}{2} + 3(x_{rs} + x_{rs}^{-1}), \quad b_{rs} = \sqrt{x_{rs}}, \quad x_{rs} = \frac{rs - 1 + 4h + \sqrt{(r-s)^2 + 8(rs-1)h + 16h^2}}{1 - r^2}. \quad (3.26)$$

At a pole c_{rs} on the complex c -plane, the residues are contributed by the descendants of this null state, which we will call $\chi_{rs} = \sum_M \chi_{rs}^M \mathbf{L}_{-M} \phi$. The pole of $\mathbf{F}_a$ at $c = c_{rs}(h_1)$ is given by

$$\begin{aligned} \mathbf{W}_{a,rs}^{(1)} = & -\frac{\partial c_{rs}(h_1)}{\partial h_1} \lim_{h_1 \rightarrow h_{r,s}} \frac{h_1 - h_{rs}}{\|\sum_M \chi_{rs}^M \mathbf{L}_{-M} \phi_1\|^2} \\ & \times \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F|, (-1)^{A+C+E}=(-1)^{a+rs} \\ (-1)^{(A+rs)C+(A+rs)E+CE}}} q_1^{|A|+\frac{rs}{2}} \eta_1^{A+rs} q_2^{|C|} \eta_2^C q_3^{|E|} \eta_3^E \mathbf{G}_{h_1+\frac{rs}{2}}^{AB} \mathbf{G}_{h_2}^{CD} \mathbf{G}_{h_3}^{EF} \\ & \rho_a(\mathbf{L}_{-A}\chi_{rs}, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3) \rho_a(\mathbf{L}_{-B}\chi_{rs}, \mathbf{L}_{-D}\phi_2, \mathbf{L}_{-F}\phi_3). \end{aligned} \quad (3.27)$$

Normalizing χ_{rs} by taking the coefficient of $G_{-1/2}^{rs} \phi$ to be 1, the inverse residue of the null norm is given by

$$\mathbf{A}_{rs}^c(h) = \lim_{h_1 \rightarrow h_{r,s}} \frac{h_1 - h_{rs}}{\|\sum_M \chi_{rs}^M \mathbf{L}_{-M} \phi_1\|^2} = \frac{1}{2} \prod_{\substack{p=1-r, \dots, r \\ q=1-s, \dots, s \\ p+q \in 2\mathbb{Z} \\ (p,q) \neq (0,0), (r,s)}} \left[\frac{pb_{rs} + qb_{rs}^{-1}}{\sqrt{2}} \right]^{-1} \quad (3.28)$$

The 3-point functions can be factorized as

$$\rho_a(\mathbf{L}_{-A}\chi_{rs}, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3) = (-1)^{rsE} \begin{cases} \rho_a(\mathbf{L}_{-A}\phi_{h_{rs}+\frac{rs}{2}}, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3) \mathbf{P}_{rs}^a(h_2, h_3), & rs \in 2\mathbb{Z}, \\ \rho_{1-a}(\mathbf{L}_{-A}\phi_{h_{rs}+\frac{rs}{2}}, \mathbf{L}_{-B}\phi_2, \mathbf{L}_{-C}\phi_3) \mathbf{P}_{rs}^a(h_2, h_3), & rs \in 2\mathbb{Z} + 1, \end{cases} \quad (3.29)$$

where the two fusion polynomials are given by

$$\begin{aligned}\mathbf{P}_{rs}^0(h_i, h_j; c) &= \prod_{\substack{p=1-r, 1-r+2, \dots, r-1 \\ q=1-s, 1-s+2, \dots, s-1 \\ p+q-(r+s) \in 4\mathbb{Z}+2}} \frac{\lambda_i - \lambda_j + pb + qb^{-1}}{2\sqrt{2}} \frac{\lambda_i + \lambda_j + pb + qb^{-1}}{2\sqrt{2}}, \\ \mathbf{P}_{rs}^1(h_i, h_j; c) &= \prod_{\substack{p=1-r, 1-r+2, \dots, r-1 \\ q=1-s, 1-s+2, \dots, s-1 \\ p+q-(r+s) \in 4\mathbb{Z}}} \frac{\lambda_i - \lambda_j + pb + qb^{-1}}{2\sqrt{2}} \frac{\lambda_i + \lambda_j + pb + qb^{-1}}{2\sqrt{2}}.\end{aligned}\tag{3.30}$$

Here λ_i is defined by

$$h_i = \frac{(b + b^{-1})^2 - \lambda_i^2}{8}.\tag{3.31}$$

Plugging these results in the pole $\mathbf{W}_{a,rs}^{(1)}$, one then have

$$\begin{aligned}\mathbf{W}_{a,rs}^{(1)} &= -\frac{\partial c_{rs}(h_1)}{\partial h_1} \mathbf{A}_{rs}^c(h) q_1^{rs/2} \eta_1^{rs} [\mathbf{P}_{rs}^a(h_2, h_3)]^2 \\ &\times \begin{cases} \mathbf{F}_a(h_1 \rightarrow h_1 + \frac{rs}{2}, c \rightarrow c_{rs}), & rs \in 2\mathbb{Z}, \\ \mathbf{F}_{1-a}(h_1 \rightarrow h_1 + \frac{rs}{2}, c \rightarrow c_{rs}, \eta_{2,3} \rightarrow -\eta_{2,3}), & rs \in 2\mathbb{Z} + 1, \end{cases}\end{aligned}\tag{3.32}$$

where the additional shift $\eta_{2,3} \rightarrow -\eta_{2,3}$ comes from the additional factor $(-1)^{rs(C+E)}$.

The next ingredient is the large- c seed term. The subalgebra of $\mathcal{N} = 1$ SCA that's independent of c is given by the $\mathfrak{osp}(1|2)$ algebra generated by $\{L_{0,\pm 1}, G_{\pm 1/2}\}$. We thus define $\hat{A}$ as the number of non- $L_{-1}, G_{-1/2}$ elements in $\mathbf{L}_{-A}$. The same c -power counting argument works also in the SUSY case: If we construct a orthonormal basis $\mathcal{L}_{-A}$ by applying Gram-Schmidt on $\mathbf{L}_{-A}$, then in the large c limit the 3-point function $\rho(\mathcal{L}_{-A}\phi_1, \mathcal{L}_{-B}\phi_2, \mathcal{L}_{-C}\phi_3)$ factorizes into the large- c vacuum block and the $\mathfrak{osp}(1|2)$ block,

$$\lim_{c \rightarrow \infty} \mathbf{F}_a(h_1, h_2, h_3, c) = \sum_{\substack{a_1, a_2, a_3 \in \{0,1\} \\ a_1 + a_2 + a_3 \equiv a \pmod{2}}} \mathbf{F}_0(0, 0, 0, \infty; \eta_i \rightarrow (-1)^{a_i} \eta_i) \times \mathbf{F}_{a_1 a_2 a_3}^{\mathfrak{osp}(1|2)}(h_1, h_2, h_3),\tag{3.33}$$

where the $\mathfrak{osp}(1|2)$ block is defined for $a_i \in \{0, 1\}$ and $a_1 + a_2 + a_3 \equiv a \pmod{2}$ by

$$\begin{aligned}\mathbf{F}_{a_1 a_2 a_3}^{\mathfrak{osp}(1|2)}(h_1, h_2, h_3) &= \sum_{k_1, k_2, k_3} q_1^{k_1 + \frac{1}{2}a_1} \eta_1^{a_1} q_2^{k_2 + \frac{1}{2}a_2} \eta_2^{a_2} q_3^{k_3 + \frac{1}{2}a_3} \eta_3^{a_3} (-1)^{a_1 a_2 + a_1 a_3 + a_2 a_3} \\ &\times \left[\prod_{i=1}^3 \frac{1}{\|L_{-1}^{k_i} G_{-\frac{1}{2}}^{a_i} \phi_i\|^2} \right] \rho_a(L_{-1}^{k_1} G_{-\frac{1}{2}}^{a_1} \phi_1, L_{-1}^{k_2} G_{-\frac{1}{2}}^{a_2} \phi_2, L_{-1}^{k_3} G_{-\frac{1}{2}}^{a_3} \phi_3)^2\end{aligned}\tag{3.34}$$

Importantly, the large- c limit of the odd vacuum block is 0, due to the fact that one can only contract G_r, G_{-r} pairwise. Due to the Grassmann parity of $G_{-1/2}^a \phi$, for each fixed a one must

sum over the corresponding 4 configurations of global $\mathfrak{sl}(2)$ primaries independently; the two values of a give 8 configurations in total. The sign η_i in the corresponding global block must be shifted. Similar to the Virasoro case, the vacuum block can also be directly computed in Schottky parameters. We will focus on the global $\mathfrak{osp}(1|2)$ blocks in the next section.

4 Global $\mathfrak{osp}(1|2)$ blocks

In this section, we develop the method of computing arbitrary Global $\mathfrak{osp}(1|2)$ blocks. We take the following basis for states

$$L_{-1}^k G_{-\frac{1}{2}}^a \phi, \quad k \in \mathbb{N}, \quad a \in \{0, 1\}. \quad (4.1)$$

with ϕ normalized such that $\|\phi\|^2 = 1$. The nontrivial inner products are

$$\|L_{-1}^k \phi\|^2 = k!(2h)_k, \quad \|L_{-1}^k G_{-\frac{1}{2}} \phi\|^2 = k!(2h)_{k+1}, \quad (4.2)$$

where $(x)_n$ is the rising Pochhammer symbol.

For the genus-2 block in the theta channel, fix $a_i \in \{0, 1\}$ and set $a \equiv a_1 + a_2 + a_3 \pmod{2}$. We then have

$$\begin{aligned} \mathbf{F}_{a_1 a_2 a_3}^{\mathfrak{osp}(1|2)}(h_1, h_2, h_3) &= \sum_{k_1, k_2, k_3} q_1^{k_1 + \frac{1}{2}a_1} \eta_1^{a_1} q_2^{k_2 + \frac{1}{2}a_2} \eta_2^{a_2} q_3^{k_3 + \frac{1}{2}a_3} \eta_3^{a_3} (-1)^{a_1 a_2 + a_1 a_3 + a_2 a_3} \\ &\times \left[\prod_{i=1}^3 \frac{1}{k_i! (2h_i)_{k_i + a_i}} \right] \rho_a(L_{-1}^{k_1} G_{-\frac{1}{2}}^{a_1} \phi_1, L_{-1}^{k_2} G_{-\frac{1}{2}}^{a_2} \phi_2, L_{-1}^{k_3} G_{-\frac{1}{2}}^{a_3} \phi_3)^2 \end{aligned} \quad (4.3)$$

The remaining object to compute is simply $\rho_a(L_{-1}^{k_1} G_{-\frac{1}{2}}^{a_1} \phi_1, L_{-1}^{k_2} G_{-\frac{1}{2}}^{a_2} \phi_2, L_{-1}^{k_3} G_{-\frac{1}{2}}^{a_3} \phi_3)$, which can be computed directly since $G_{-\frac{1}{2}}^a \phi$ is a $\mathfrak{sl}(2)$ primary and one can use the result for $\mathfrak{sl}(2)$ 3-point functions. To convert the component coefficients obtained from the superspace expansion to the fixed-parity trilinear form used here, one must include the factor $(-1)^{a_2(1-a_1)}$. Using the superspace formula (3.3) with this conversion, one can extract

$$\frac{(a_1 a_2 a_3)}{\rho_0(G_{-1/2}^{a_1} \phi_1, G_{-1/2}^{a_2} \phi_2, G_{-1/2}^{a_3} \phi_3)} \left| \begin{array}{cccc} 000 & 110 & 101 & 011 \\ 1 & h_1 + h_2 - h_3 & h_1 - h_2 + h_3 & h_1 - h_2 - h_3 \end{array} \right. \quad (4.4)$$

For the parity-odd correlators,

$$\frac{(a_1 a_2 a_3)}{\rho_1(G_{-1/2}^{a_1} \phi_1, G_{-1/2}^{a_2} \phi_2, G_{-1/2}^{a_3} \phi_3)} \left| \begin{array}{cccc} 100 & 010 & 001 & 111 \\ 1 & 1 & -1 & -(h_1 + h_2 + h_3 - \frac{1}{2}) \end{array} \right. \quad (4.5)$$

Defining

$$s_{km}(h_1, h_2, h_3) = \sum_{p=0}^{\min(k,m)} \frac{k!}{p!(k-p)!} (2h_3 + m - 1)^{(p)} m^{(p)} \\ \times (h_3 + h_2 - h_1)_{m-p} (h_1 + h_2 - h_3 + p - m)_{k-p},$$

where $(x)^{(n)}$ is the falling Pochhammer symbol. Using the formula for $\mathfrak{sl}(2)$ 3-point functions in [Chollier-Yin], one then arrive at

$$\rho_a \left(L_{-1}^{k_1} G_{-1/2}^{a_1} \phi_1, L_{-1}^{k_2} G_{-1/2}^{a_2} \phi_2, L_{-1}^{k_3} G_{-1/2}^{a_3} \phi_3 \right) \\ = \rho_a \left(G_{-1/2}^{a_1} \phi_1, G_{-1/2}^{a_2} \phi_2, G_{-1/2}^{a_3} \phi_3 \right) s_{k_1 k_3} \left(h_1 + \frac{a_1}{2}, h_2 + \frac{a_2}{2}, h_3 + \frac{a_3}{2} \right) \\ \times \left(h_1 + \frac{a_1}{2} + k_1 - h_2 - \frac{a_2}{2} - k_2 + 1 - h_3 - \frac{a_3}{2} - k_3 \right)_{k_2}, \quad (4.6)$$

and hence

$$\mathbf{F}_{a_1 a_2 a_3}^{\text{osp}(1|2)}(h_1, h_2, h_3) = \sum_{k_1, k_2, k_3} q_1^{k_1 + \frac{1}{2} a_1} \eta_1^{a_1} q_2^{k_2 + \frac{1}{2} a_2} \eta_2^{a_2} q_3^{k_3 + \frac{1}{2} a_3} \eta_3^{a_3} (-1)^{a_1 a_2 + a_1 a_3 + a_2 a_3} \left[\prod_{i=1}^3 \frac{1}{k_i! (2h_i)_{k_i + a_i}} \right] \\ \times \left[\rho_a \left(G_{-1/2}^{a_1} \phi_1, G_{-1/2}^{a_2} \phi_2, G_{-1/2}^{a_3} \phi_3 \right) s_{k_1 k_3} \left(h_1 + \frac{a_1}{2}, h_2 + \frac{a_2}{2}, h_3 + \frac{a_3}{2} \right) \right. \\ \left. \times \left(h_1 + \frac{a_1}{2} + k_1 - h_2 - \frac{a_2}{2} - k_2 + 1 - h_3 - \frac{a_3}{2} - k_3 \right)_{k_2} \right]^2. \quad (4.7)$$

Using this formula, one can compute the $\text{osp}(1|2)$ block directly as a q -expansion. This concludes our discussion for the all-NS c -recursion relation.

5 The Ramond blocks

Let us now consider the case with intermediate Ramond supermultiplets. The h -recursion relation of sphere 4-point Ramond block is derived in [0810.1203, 1012.2974], and that of torus 1-point Ramond block is derived in [1207.5740]. In this section, we aim for c -recursion of arbitrary Ramond blocks.

A Ramond multiplet has 2 ground states $w^\pm$ that are annihilated by $L_{m>0}, G_{n>0}$ and carries L_0 eigenvalue h . They are related by acting G_0 (we use the convention of [0810.1203] for R-supermultiplet):

$$G_0 w^\pm = i\beta e^{\mp i\pi/4} w^\mp, \quad (5.1)$$

where β is defined by $\beta^2 = \frac{c}{24} - h$. For R-sector c -recursion relation, instead of fixing the internal weight h , we fix β of the internal primary, as doing so keeps the G_0 action on $w^\pm$ invariant. We take w^+ to be Grassmann even, and hence w^- will be Grassmann odd. An arbitrary state in the Ramond multiplet can be then written as

$$\mathbb{L}_{-A} w^\pm \equiv L_{-n_1} \cdots L_{-n_k} G_{-m_1} \cdots G_{-m_\ell} w^\pm, \quad |A| = \sum_{i=1}^k n_i + \sum_{j=1}^\ell m_j, \quad (-1)^A \equiv (-1)^\ell, \quad (5.2)$$

where $n_1 \geq \cdots \geq n_k, m_1 > \cdots > m_\ell$. Throughout this notes, the NS-sector quantities will be written in **mathbf{b}**, and the R-sector quantities in **mathbf{bb}**.

For a sphere with (NS, R, R) insertions, we normalize the 3-point function such that

$$\rho^{++}(\phi_1, w_2^+, w_3^+) = 1, \quad \rho^{+-}(\phi_1, w_2^+, w_3^-) = 1, \quad \rho^{-+}(\phi_1, w_2^-, w_3^+) = 1, \quad \rho^{--}(\phi_1, w_2^-, w_3^-) = 1. \quad (5.3)$$

The 3-point functions with other slot orders can be related to these basic ones by a Mobius transformation. It turns out that all $\rho^{\pm, \pm}$'s with all slot orders are normalized to 1. One can introduce [1012.2974]

$$\begin{aligned} \rho_0^{(\eta)}(\xi_1, \xi_2, \xi_3) &= \rho^{++}(\xi_1, \xi_2, \xi_3) + \eta \rho^{--}(\xi_1, \xi_2, \xi_3), \\ \rho_1^{(\eta)}(\xi_1, \xi_2, \xi_3) &= \rho^{+-}(\xi_1, \xi_2, \xi_3) + i\eta \rho^{-+}(\xi_1, \xi_2, \xi_3), \end{aligned} \quad (5.4)$$

for $\eta = \pm$. The subscript 0, 1 indicates the total fermion parity, as $\rho(\mathbf{L}_{-A}\phi, \mathbb{L}_{-C}w^\alpha, \mathbb{L}_{-E}w^\gamma)$ can eventually be reduced to $\rho(\phi, w^{\alpha'}, w^{\gamma'})$ with $(-1)^{|\alpha'|+|\gamma'|} = (-1)^{A+C+E+|\alpha'|+|\gamma'|}$, where $|\alpha| = 0$ for $\alpha = +$ and $|\alpha| = 1$ for $\alpha = -$, as the total fermion parity stays the same under the contour pulling used in Ward identities.

The Ramond block in the glasses channel, with spin structure specified by the three signs (η_1, η_2, η_3) and the edge 1 being NS, is defined by

$$\begin{aligned} \mathbb{F}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c) &= \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F| \\ (-1)^{A+C+E+|\alpha|+|\gamma|} = (-1)^f \\ (-1)^{B+D+F+|\alpha'|+|\gamma'|} = (-1)^f}} q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^{C+|\alpha|} q_3^{|E|} \eta_3^{E+|\gamma|} \mathbf{G}_{h_1}^{AB} \mathbb{G}_{\beta_2}^{C\alpha, D\alpha'} \mathbb{G}_{\beta_3}^{E\gamma, F\gamma'} \\ &\quad \times (-1)^{A(C+|\alpha|)+A(E+|\gamma|)+(C+|\alpha|)(E+|\gamma|)} \\ &\quad \times \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-C}w_2^\alpha, \mathbb{L}_{-E}w_3^\gamma) \rho_f^{(\eta')}(\mathbf{L}_{-B}\phi_1, \mathbb{L}_{-D}w_2^{\alpha'}, \mathbb{L}_{-F}w_3^{\gamma'}) \end{aligned} \quad (5.5)$$

where $\mathbb{G}$ is the inverse Gram matrix for R-multiplets (which excludes actions of G_0), and $\alpha, \alpha', \gamma, \gamma'$ are such that the total fermion parity $(-1)^{A+C+E+|\alpha|+|\gamma|} = (-1)^f$.

Let's consider the case where the w_2 multiplet develops a null descendant. This happens when

$$\beta = \beta_{rs}(c) = \frac{rb + sb^{-1}}{2\sqrt{2}}, \quad r + s \in 2\mathbb{Z} + 1 \quad (5.6)$$

and for fixed β , this becomes the following value for c :

$$\begin{aligned} x_{rs}^R(\beta) &= \frac{4\beta^2 - rs + 2\sqrt{2}\beta\sqrt{2\beta^2 - rs}}{r^2}, \\ c_{rs}^R(\beta) &= \frac{15}{2} + 3(x_{rs}^R(\beta) + x_{rs}^R(\beta)^{-1}). \end{aligned} \quad (5.7)$$

We choose $b_{rs} = \sqrt{x_{rs}^R}$ continuously such that $\beta = \beta_{rs}(c_{rs}^R)$. Below, b_{rs} and c_{rs}^R refer to this branch. When this happens, the R multiplet develops a doublet of null states $\chi_{rs}^\pm = \sum_{M,\alpha} \chi_{rs}^{M,\alpha} G_0^\alpha \mathbb{L}_{-M} w^\pm$ happens at level $h + rs/2$, and we normalize $\chi_{rs}^\pm$ such that the coefficient of $L_{-1}^{rs/2} w^\pm$ is 1. In β variables, the weight $h_{rs} = h + rs/2$ translates to

$$\beta'_{rs} = \frac{(-1)^s}{2\sqrt{2}} (rb_{rs} - sb_{rs}^{-1}). \quad (5.8)$$

The Ramond block $\mathbb{F}_f^{(\eta,\eta')}(h_1, \beta_2, \beta_3, c)$ develops a pole when $c = c_{rs}^R(\beta_2)$. The inverse norm for the null state is given by

$$\mathbb{A}_{rs}^c \equiv \lim_{c \rightarrow c_{rs}^R(\beta)} \frac{\beta_{rs}(c)^2 - \beta^2}{\|\chi_{rs}^+\|^2} = 2^{rs-2} (rb_{rs} + sb_{rs}^{-1}) \prod_{\substack{p=1-r,\dots,r, q=1-s,\dots,s \\ p+q \in \mathbb{Z}, (p,q) \neq (0,0)}} (pb_{rs} + qb_{rs}^{-1})^{-1} \quad (5.9)$$

and the fusion polynomials are given by

$$\begin{aligned} \mathbb{P}_{rs}^{R,\eta}(h_i, \beta_j; c) &= \prod_{\substack{0 \leq k \leq r-1, 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}}} \frac{\lambda_i - 2\sqrt{2}\eta\beta_j - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}} \\ &\times \prod_{\substack{0 \leq k \leq r-1, 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}+1}} \frac{\lambda_i + 2\sqrt{2}\eta\beta_j - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}}. \end{aligned} \quad (5.10)$$

The factorization relations are summarized in appendix A. Using the relation

$$\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}\chi_{rs}^\alpha, \mathbb{L}_{-E}w_{\beta_j}^\gamma) = \mathbb{P}_{rs}^{R,\eta}(h_i, \beta_j; c_{rs}^R)\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}w_{\beta'_{rs}}^\alpha, \mathbb{L}_{-E}w_{\beta_j}^\gamma) \quad (5.11)$$

one can read off the polar value of $\mathbb{F}_f^{(\eta,\eta')}$ at $c = c_{rs}^R(\beta_2)$:

$$\begin{aligned} \text{Res}_{c=c_{rs}^R(\beta_2)} \mathbb{F}_f^{(\eta,\eta')} &= \frac{\partial c_{rs}(\beta_2)}{\partial \beta_2^2} \mathbb{A}_{rs}^c \sum_{\substack{|A|=|B|, |C|=|D|, |E|=|F| \\ (-1)^{A+C+E+|\alpha|+|\gamma|} = (-1)^f \\ (-1)^{B+D+F+|\alpha'|+|\gamma'|} = (-1)^f}} q_1^{|A|} \eta_1^A q_2^{|C|} \eta_2^{C+|\alpha|} q_3^{|E|} \eta_3^{E+|\gamma|} \\ &\times (-1)^{A(C+|\alpha|)+A(E+|\gamma|)+(C+|\alpha|)(E+|\gamma|)} \mathbf{G}_{h_1}^{AB} \mathbb{G}_{\beta'_{rs}}^{C\alpha, D\alpha'} \mathbb{G}_{\beta_3}^{E\gamma, F\gamma'} \\ &\times \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-C}\chi_{rs}^\alpha, \mathbb{L}_{-E}w_3^\gamma) \rho_f^{(\eta')}(\mathbf{L}_{-B}\phi_1, \mathbb{L}_{-D}\chi_{rs}^\alpha, \mathbb{L}_{-F}w_3^{\gamma'}), \end{aligned} \quad (5.12)$$

and hence

$$\begin{aligned} \text{Res}_{c=c_{rs}^R(\beta_2)} \mathbb{F}_f^{(\eta, \eta')} &= \frac{\partial c_{rs}(\beta_2)}{\partial \beta_2^2} \mathbb{A}_{rs}^c \mathbb{P}_{rs}^{R, \eta}(h_1, \beta_3; c_{rs}^R(\beta_2)) \mathbb{P}_{rs}^{R, \eta'}(h_1, \beta_3; c_{rs}^R(\beta_2)) \\ &\times \mathbb{F}_f^{(\eta, \eta')}(\beta_2 \rightarrow \beta'_{rs}, c \rightarrow c_{rs}^R(\beta_2)) \end{aligned} \quad (5.13)$$

where we used the fact that $\chi_{rs}^{M, \alpha} G_0^\alpha \mathbb{L}_{-M}$ is Grassmann even, which ensures that the total fermion parity f doesn't get shifted and there are no additional signs η_2 . The residue at $c = c_{rs}^R(\beta_3)$ can be worked out analogously.

The Ramond also develops poles when the NS supermultiplet contains a null state. The relevant fusion polynomials are

$$\begin{aligned} \mathbb{P}_{rs}^{\text{NS}, \eta}(\beta_2, \beta_3; c) &= \prod_{\substack{0 \leq k \leq r-1, 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}}} \frac{2\sqrt{2}(\beta_3 - \eta\beta_2) - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}} \\ &\times \prod_{\substack{0 \leq k \leq r-1, 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}+1}} \frac{2\sqrt{2}(\beta_3 + \eta\beta_2) - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}}, \end{aligned} \quad (5.14)$$

and the factorization relation is given by

$$\begin{aligned} \rho_f^{(\eta)}(\mathbb{L}_{-A}\chi_{rs}, \mathbb{L}_{-C}w_{\beta_2}^\alpha, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \\ = \mathbb{P}_{rs}^{\text{NS}, \eta}(\beta_2, \beta_3; c_{rs}) \begin{cases} \rho_f^{(\eta)}(\mathbb{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-C}w_{\beta_2}^\alpha, \mathbb{L}_{-E}w_{\beta_3}^\gamma), & rs \in 2\mathbb{Z}, \\ e^{i(-1)^f \pi/4} \rho_{1-f}^{(-\eta)}(\mathbb{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-C}w_{\beta_2}^\alpha, \mathbb{L}_{-E}w_{\beta_3}^\gamma), & rs \in 2\mathbb{Z}+1. \end{cases} \end{aligned} \quad (5.15)$$

From these, one can derive the residue at $c = c_{rs}(h_1)$:

$$\begin{aligned} \text{Res}_{c=c_{rs}(h_1)} \mathbb{F}_f^{(\eta, \eta')} &= -\frac{\partial c_{rs}(h_1)}{\partial h_1} \mathbb{A}_{rs}^c(h_1) q_1^{rs/2} \mathbb{P}_{rs}^{\text{NS}, \eta}(\beta_2, \beta_3; c_{rs}(h_1)) \mathbb{P}_{rs}^{\text{NS}, \eta'}(\beta_2, \beta_3; c_{rs}(h_1)) \\ &\times \begin{cases} \mathbb{F}_f^{(\eta, \eta')}(h_1 \rightarrow h_1 + \frac{rs}{2}, c \rightarrow c_{rs}(h_1)), & rs \in 2\mathbb{Z}, \\ i(-1)^f \eta_1 \mathbb{F}_{1-f}^{(-\eta, -\eta')}(h_1 \rightarrow h_1 + \frac{rs}{2}, c \rightarrow c_{rs}(h_1), \eta_2 \rightarrow -\eta_2, \eta_3 \rightarrow -\eta_3), & rs \in 2\mathbb{Z}+1. \end{cases} \end{aligned} \quad (5.16)$$

We now have computed all the polar part in the following recursion relation

$$\begin{aligned} \mathbb{F}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c) &= \mathbb{F}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c \rightarrow \infty) \\ &+ \sum_{r+s \in 2\mathbb{Z}} \frac{\text{Res}_{c=c_{rs}(h_1)} \mathbb{F}_f^{(\eta, \eta')}}{c - c_{rs}(h_1)} + \sum_{i=2,3} \sum_{r+s \in 2\mathbb{Z}+1} \frac{\text{Res}_{c=c_{rs}^R(\beta_i)} \mathbb{F}_f^{(\eta, \eta')}}{c - c_{rs}^R(\beta_i)} \end{aligned} \quad (5.17)$$

The next ingredient of the c -recursion relation of the Ramond blocks is the large- c limit, which we will compute in the next section.

6 The large- c Ramond block

In the $c \rightarrow \infty$ limit, we fix h of the NS supermultiplet, and fix β of the R supermultiplet, which implies that $L_0 = \frac{c}{24} - \beta^2 + |A|$ is of order $\mathcal{O}(c)$ for a Ramond state $\mathbb{L}_{-A} w_\beta^\pm$.

We first analyze the c -scaling behavior of the Gram matrix element $\langle w_\beta^\alpha | \mathbb{L}_{-A}^\dagger \mathbb{L}_{-B} | w_\beta^\gamma \rangle$, where there are no G_0 's in the lowering operators $\mathbb{L}_{-A}$. The SCA in the R-sector reads

$$\begin{aligned} [L_m, L_n] &= (m-n)L_{m+n} + \frac{c}{12} (m^3 - m) \delta_{m+n,0}, \\ [L_m, G_n] &= \left(\frac{m}{2} - n\right) G_{m+n}, \\ \{G_m, G_n\} &= 2L_{m+n} + \frac{c}{3} \left(m^2 - \frac{1}{4}\right) \delta_{m+n,0}. \end{aligned} \quad (6.1)$$

Therefore, in the large- c limit, the algebra becomes

$$[L_m, L_{-n}] = \frac{c}{12} m^3 \delta_{m,n}, \quad \{G_m, G_{-n}\} = \frac{c}{3} m^2 \delta_{m,n}, \quad [L_m, G_{-n}] = 0. \quad (6.2)$$

The leading term in the Gram matrix then comes from pairwise contractions between the generators. Let $\hat{A}$ be the total number of generators in $\mathbb{L}_{-A}$, then $\langle w_\beta^\alpha | \mathbb{L}_{-A}^\dagger \mathbb{L}_{-B} | w_\beta^\gamma \rangle$ is of order $\mathcal{O}(c^{\hat{A}})$ if $(A, \alpha) = (B, \gamma)$, at most $\mathcal{O}(c^{\hat{A}-1})$ if $\hat{A} = \hat{B}$, $(A, \alpha) \neq (B, \gamma)$, and at most $\mathcal{O}(c^{\min(\hat{A}, \hat{B})})$ if $\hat{A} \neq \hat{B}$. One can then apply Gram-Schmidt to $\mathbb{L}_{-A} w_\beta^\pm$ to obtain a orthonormal basis $\mathcal{L}_{-A} w_\beta^\pm$, and

$$\|\mathcal{L}_{-A} w_\beta^\pm\|^2 \sim \mathcal{O}(c^{\hat{A}}). \quad (6.3)$$

The next ingredient is the scaling behavior of the 3-point function $\rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma)$. One can use conformal Ward identities to convert a L_{-n}, G_{-r} insertion in one slot to another slot. For a L_{-n} insertion,

$$\begin{aligned} &\rho_f^{(\eta)}(L_{-n} \mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma) \\ &= \rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, L_n \mathbb{L}_{-C} w_3^\gamma) + \sum_{p=-1}^n \binom{n+1}{p+1} \rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, L_p \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma), \\ &\rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, L_{-n} \mathbb{L}_{-C} w_3^\gamma) \\ &= \rho_f^{(\eta)}(L_n \mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma) - \sum_{p=0}^\infty \binom{1-n}{p} \rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, L_{p-1} \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma), \\ &\rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, L_{-n} \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma) = \sum_{p=0}^\infty \binom{n-2+p}{n-2} \rho_f^{(\eta)}(L_{n+p} \mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma) \\ &\quad + (-1)^n \sum_{p=0}^\infty \binom{n-2+p}{n-2} \rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, L_{p-1} \mathbb{L}_{-C} w_3^\gamma). \end{aligned} \quad (6.4)$$

This imposes a problem to our infinite- c limit. Each conformal Ward identity contains a L_0 term acting on a Ramond multiplet on the RHS. In our fixed β large- c limit, this will contribute to a term of order $\mathcal{O}(c)$. Therefore, for $\mathbf{L}_{-A}, \mathbb{L}_{-A}$ made from purely L_{-n} 's,

$$\rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_1, \mathbb{L}_{-B}w_2^\alpha, \mathbb{L}_{-C}w_3^\gamma) \sim \mathcal{O}(c^{\hat{A}+\hat{B}+\hat{C}}) \quad (6.5)$$

and the contribution of this term to the large- c conformal block will scale like $\mathcal{O}(c^{\hat{A}+\hat{B}+\hat{C}})$. Namely, the large- c , fixed- β limit of the Ramond block is not well-defined.

7 NS blocks from double Virasoro

We now turn to another approach of computing the superconformal blocks. The idea is: Given the $\mathcal{N} = 1$ SCA with shift $\nu \in \{0, \frac{1}{2}\}$ in the fermionic level, one can construct an enlarged algebra $\text{SCA}_\nu \otimes \text{F}_\nu$, where F_ν a auxiliary free fermion algebra with the same level shift ν . $\text{SCA}_\nu \otimes \text{F}_\nu$ admits two copy of virasoro algebra as it's subalgebra $\text{Vir} \oplus \text{Vir} \subset \text{SCA}_\nu \otimes \text{F}_\nu$ [1312.4520, 1608.02568]. Using the branching rules of this subalgebra reduction, one can decompose the superconformal block into ordinary Virasoro blocks.

The enlarged algebra $\text{SCA}_\nu \otimes \text{F}_\nu$ is given by:

$$\begin{aligned} [L_m, L_n] &= (m-n)L_{m+n} + \frac{c}{12}(m^3-m)\delta_{m+n,0}, \\ [L_m, G_r] &= \left(\frac{m}{2}-r\right)G_{m+r}, \\ \{G_r, G_s\} &= 2L_{r+s} + \frac{c}{3}\left(r^2-\frac{1}{4}\right)\delta_{r+s,0}, \\ \{\psi_r, \psi_s\} &= \delta_{r+s,0}, \quad \{\psi_r, G_s\} = [\psi_r, L_n] = 0. \end{aligned} \quad (7.1)$$

where the levels $m, n \in \mathbb{Z}$, $r, s \in \mathbb{Z} + \nu$. The reality conditions are

$$\psi_r^\dagger = -\psi_{-r}, \quad G_r^\dagger = G_{-r}, \quad L_n^\dagger = L_{-n}. \quad (7.2)$$

We also introduce the following operators

$$L_n^{\text{F}} = \frac{1}{2} \sum_{r \in \mathbb{Z} + \nu} r : \psi_{n-r} \psi_r : + \frac{1-2\nu}{16} \delta_{n,0}, \quad U_n = \sum_{r \in \mathbb{Z} + \nu} \psi_{n-r} G_r. \quad (7.3)$$

Define b by the relation

$$c = \frac{3}{2} + 3(b+b^{-1})^2. \quad (7.4)$$

For $b \neq 1$, one can then define

$$\begin{aligned} L_n^{(1)} &= \frac{b^{-1}}{b^{-1}-b} L_n - \frac{b^{-1}+2b}{b^{-1}-b} L_n^{\text{F}} + \frac{1}{b^{-1}-b} U_n, \\ L_n^{(2)} &= \frac{b}{b-b^{-1}} L_n - \frac{b+2b^{-1}}{b-b^{-1}} L_n^{\text{F}} + \frac{1}{b-b^{-1}} U_n, \end{aligned} \quad (7.5)$$

which satisfies $L_n^{(1)} + L_n^{(2)} = L_n + L_n^F$. Note that in order to have $(L_n^{(i)})^\dagger = L_{-n}^{(i)}$, we need to take the BPZ conjugate of ψ_r to be $(\psi_r)^\dagger = -\psi_{-r}$. One can also prove that these operators are the generators of two commuting copies of the Virasoro algebra

$$[L_m^{(i)}, L_n^{(j)}] = \left[(m-n)L_{m+n}^{(i)} + \frac{c^{(i)}}{12}(m^3-m)\delta_{m+n,0} \right] \delta_{ij}, \quad (7.6)$$

and the two central charges are given by

$$c^{(i)} = 1 + 6 \left(b^{(i)} + \frac{1}{b^{(i)}} \right)^2, \quad (b^{(1)})^2 = \frac{2b^2}{1-b^2}, \quad (b^{(2)})^{-2} = \frac{2}{b^2-1}. \quad (7.7)$$

This proves the subalgebra relation $\text{Vir} \oplus \text{Vir} \subset \text{SCA}_\nu \otimes \text{F}_\nu$.

Under the subalgebra reduction $\text{Vir} \oplus \text{Vir} \subset \text{SCA}_\nu \otimes \text{F}_\nu$, a highest-weight representation of $\text{SCA}_\nu \otimes \text{F}_\nu$ can be decomposed into irreps of $\text{Vir} \oplus \text{Vir}$. This decomposition is worked out in [1608.02568]. We now discuss this decomposition in both NS and R sectors.

NS Sector

As always, let $\mathcal{V}_{\text{SCA}_{1/2}}(h)$ denote the NS Verma module generated by an highest-weight state ϕ of conformal weight h . Let $\mathcal{V}_{F_{1/2}}$ denote the Fock representation of $F_{1/2}$. The representation of interest is $\mathcal{M}_{\text{NS}}(h) := \mathcal{V}_{F_{1/2}} \otimes \mathcal{V}_{\text{SCA}_{1/2}}(h)$. It can be generated by acting arbitrary $\psi_{-r}, L_{-n}, G_{-r}$ on $\mathbf{1}_F \otimes \phi$.

We further define the "Liouville momentum" P by

$$h = \frac{1}{2} \left(\frac{1}{4} Q^2 - P^2 \right), \quad (7.8)$$

where $Q = b + b^{-1}$. Denoting the Virasoro Verma module with lowest weight h by $\mathcal{V}_c(h)$, it's proven that [1111.2803]

$$\mathcal{M}_{\text{NS}}(h) \cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} \mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)}) \quad (7.9)$$

for $P \notin \frac{1}{2}(b\mathbb{Z} + b^{-1}\mathbb{Z})$. The weights $h_n^{(i)}$'s are given by $h_n^{(i)} = \frac{1}{4}(Q^{(i)})^2 - (P_n^{(i)})^2$, where

$$P_n^{(1)} = \frac{P}{\sqrt{2-2b^2}} + nb^{(1)}, \quad P_n^{(2)} = \frac{P}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1}. \quad (7.10)$$

It satisfies $h_n^{(1)} + h_n^{(2)} = h + 2n^2$.

We will denote the highest weight vector in $\mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)})$ by v_n . In our convention where ϕ is Grassmann even, v_n has Grassmann parity $2n$. The Liouville momentum P gives

identical $\mathcal{V}_{\text{SCA}_{1/2}}$, and a reflection $(n, P) \mapsto (-n, -P)$ gives the same $\text{Vir} \oplus \text{Vir}$ module. We can then identify $v_n(P) = v_{-n}(-P)$ and work primarily with $P > 0$. For multiplets with $h > Q^2/8$, we analytic continue all the formula from positive real P to purely imaginary $P = ip$ with $p > 0$.

The standard bilinear pairing on $\mathcal{M}_{\text{NS}}(h)$ in the theory of chiral fermion coupled to superconformal field theory is given by

$$\langle\langle \Psi_{-A} \mathbf{1}_F \otimes \mathbf{L}_{-A} \phi \mid \Psi_{-B} \mathbf{1}_F \otimes \mathbf{L}_{-B} \phi \rangle\rangle = (-1)^{\text{B}(A+p_\phi)} \langle\langle \Psi_{-A} \mathbf{1}_F \mid \Psi_{-B} \mathbf{1}_F \rangle\rangle \times \langle\langle \mathbf{L}_{-A} \phi \mid \mathbf{L}_{-B} \phi \rangle\rangle. \quad (7.11)$$

Introduce the parity operator $(-1)^{p_{\text{SCA}} \cdot p_F}$. We have

$$\langle\langle \Psi_{-A} \mathbf{1}_F \otimes \mathbf{L}_{-A} \phi \mid (-1)^{p_{\text{SCA}} \cdot p_F} \Psi_{-B} \mathbf{1}_F \otimes \mathbf{L}_{-B} \phi \rangle\rangle = \langle\langle \Psi_{-A} \mathbf{1}_F \mid \Psi_{-B} \mathbf{1}_F \rangle\rangle \times \langle\langle \mathbf{L}_{-A} \phi \mid \mathbf{L}_{-B} \phi \rangle\rangle. \quad (7.12)$$

where we use $A + B = 0 \pmod{2}$ for the matrix element to be non-vanishing.

This parity operator is useful because the inner produce of Virasoro generator would factorize with this operator insertion

$$\langle\langle \mathbf{L}_{-M}^{(1)} \nu_1 \otimes \mathbf{L}_{-N}^{(2)} \nu_2 \mid (-1)^{p_{\text{SCA}} \cdot p_F} \mathbf{L}_{-M'}^{(1)} \otimes \nu_1 \mathbf{L}_{-N'}^{(2)} \otimes \nu_2 \rangle\rangle = \langle\langle \mathbf{L}_{-M}^{(1)} \nu_1 \mid \mathbf{L}_{-M'}^{(1)} \nu_1 \rangle\rangle \langle\langle \mathbf{L}_{-N}^{(2)} \nu_2 \mid \mathbf{L}_{-N'}^{(2)} \nu_2 \rangle\rangle. \quad (7.13)$$

R Sector

Let $\mathcal{V}_{\text{SCA}_0}(\beta)$ denote the R Verma module generated by an highest-weight state w_β^+ of conformal weight $h = \frac{c}{24} - \beta^2$. Two degenerate ground states will be denoted as $w_\beta^\pm$. Let $\mathcal{V}_{F_0}$ denote the Fock representation of F_0 . Two ground states in $\mathcal{V}_{F_0}$ will be denoted as $u^\pm$, with $\psi_0 u^\pm = 2^{-1/2} u^\mp$.

The representation of interest is $\mathcal{M}_{\text{R}}(\beta) := \mathcal{V}_{F_0} \otimes \mathcal{V}_{\text{SCA}_0}(\beta)$. It contains 4 ground states given by $u^\pm \otimes w_\beta^\pm$. $\mathcal{M}_{\text{R}}(\beta)$ can be generated by acting lowering operators on the 4 ground states $u^\pm \otimes w_\beta^\pm$. The Liouville momentum of $w_\beta^\pm$ is defined by

$$h = \frac{c}{24} - \beta^2 = \frac{1}{16} + \frac{1}{2} \left(\frac{1}{4} Q^2 - P^2 \right), \quad P^2 = 2\beta^2. \quad (7.14)$$

For $P \notin \frac{1}{2}(b\mathbb{Z} + b^{-1}\mathbb{Z})$, $\mathcal{M}_{\text{R}}(\beta)$ can be decomposed into [1608.02568]

$$\mathcal{M}_{\text{R}}(\beta) \cong \bigoplus_{\epsilon=0,1} \bigoplus_{n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}} \mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)}), \quad (7.15)$$

where

$$h_n^{(i)} = \frac{1}{4}(Q^{(i)})^2 - (P_n^{(i)})^2, \quad P_n^{(1)} = \frac{P}{\sqrt{2-2b^2}} + nb^{(1)}, \quad P_n^{(2)} = \frac{P}{\sqrt{2-2b^{-2}}} + n(b^{(2)})^{-1}. \quad (7.16)$$

It satisfies $h_n^{(1)} + h_n^{(2)} = h + 2n^2 - \frac{1}{16}$.

The direct sum over ϵ indicates that there are two copies of each $\mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)})$, and ϵ denotes the total fermion parity of the lowest weight state (including the ψ parity). One can directly check that the characters of the two sides of (7.9), (7.15) agrees via the Jacobi triple product identity. The ground state of $\mathcal{V}_{c(1)}(h_n^{(1)}) \otimes \mathcal{V}_{c(2)}(h_n^{(2)})$ will be denoted by $v_n^\pm$. We use the convention that v_n^α has Grassmann parity α .

Decomposition of all-NS block

Let's inspect on how the superconformal blocks $\mathbf{F}_a(h_1, h_2, h_3, c)$, $\mathbb{F}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c)$ decompose under $\text{Vir} \oplus \text{Vir} \subset \text{SCA}_\nu \otimes \text{F}$. We first define the all-NS $\text{SCA}_\nu \otimes \text{F}$ block in the theta channel as

$$\begin{aligned} \widehat{\mathbf{F}}_a(h_1, h_2, h_3, c) = & \sum_{\substack{|A|=|A'|, |B|=|B'|, |C|=|C'|, \mathbf{A}, \mathbf{B}, \mathbf{C} \\ (-1)^{A+B+C}(-1)^{A+B+C}=(-1)^a}} q_1^{|A|} \eta_1^A q_2^{|B|} \eta_2^B q_3^{|C|} \eta_3^C \mathbf{G}_{h_1}^{AA'} \mathbf{G}_{h_2}^{BB'} \mathbf{G}_{h_3}^{CC'} \\ & \times (-1)^{(A+A)(B+B)+(A+A)(C+C)+(B+B)(C+C)} (-1)^{A+B+C} q_1^{|A|} \eta_1^A q_2^{|B|} \eta_2^B q_3^{|C|} \eta_3^C \\ & \times \hat{\rho}_a(\Psi_{-A} \mathbf{L}_{-A} \phi_1, \Psi_{-B} \mathbf{L}_{-B} \phi_2, \Psi_{-C} \mathbf{L}_{-C} \phi_3) \hat{\rho}_a(\Psi_{-A} \mathbf{L}_{-A'} \phi_1, \Psi_{-B} \mathbf{L}_{-B'} \phi_2, \Psi_{-C} \mathbf{L}_{-C'} \phi_3), \end{aligned} \quad (7.17)$$

where $\Psi_{-A} v$ stands for $\Psi_{-A} v \equiv \psi_{-r_1} \cdots \psi_{-r_m} v$, $|A|$ stands for the total level of fermion lowering operators, and $(-1)^A$ is the total ψ parity. We have used the fact that $\langle \mathbf{L}_{-A} v | \Psi_{-A}^\dagger (-1)^{p_{\text{SCA}} \cdot p_{\text{F}}} \Psi_{-B} | \mathbf{L}_{-B} v \rangle$ is the matrix $(-1)^A \delta_{A,B} \mathbf{G}_{AB}$, which give rises to the sign $(-1)^{A+B+C}$. The 3-point function $\hat{\rho}^{L/R}$ is labeled by the total parity, including the fermionic one.

In the above definition, each sum resolves the $(-1)^{p_{\text{SCA}} \cdot p_{\text{F}}}$ operator along the plumbing cycle. It should be distinguished from the genus 2 block in chiral free fermion + superconformal field theory.

For states $u \otimes x$, the 3-point function can be factorized into

$$\hat{\rho}_a(u_1 \otimes x_1, u_2 \otimes x_2, u_3 \otimes x_3) = (-1)^{|x_1|(|u_2|+|u_3|)+|x_2||u_3|} \rho^{\text{F}}(u_1, u_2, u_3) \rho_a(x_1, x_2, x_3), \quad (7.18)$$

where $\rho^{\text{F}}(u_1, u_2, u_3)$ is the 3-point function in the free fermion theory. The sign $(-1)^{|x_1|(|u_2|+|u_3|)+|x_2||u_3|}$ comes from moving operator order, and the additional sign $(-1)^{a|u_3|}$ comes from Importantly, ρ^{F} is only nonzero if $A + B + C \equiv 0 \pmod{2}$. If we define the free fermion block as

$$\mathbf{F}^{\text{F}}(q_i, \eta_i) = \sum_{\substack{\mathbf{A}, \mathbf{B}, \mathbf{C} \\ (-1)^{A+B+C}=1}} (-1)^{AB+AC+BC} q_1^{|A|} \eta_1^A q_2^{|B|} \eta_2^B q_3^{|C|} \eta_3^C [\rho^{\text{F}}(\Psi_{-A} \mathbf{1}_{\text{F}}, \Psi_{-B} \mathbf{1}_{\text{F}}, \Psi_{-C} \mathbf{1}_{\text{F}})]^2, \quad (7.19)$$

then the all-NS $\text{SCA}_\nu \otimes \text{F}$ block can be factorized into

$$\widehat{\mathbf{F}}_a(h_1, h_2, h_3, c; q_i, \eta_i) = \mathbf{F}_{\text{F}}(q_i, \eta_i) \star \mathbf{F}_a(h_1, h_2, h_3, c; q_i, \eta_i), \quad (7.20)$$

where the symbol $\star$ stands for convolution: Denote the coefficient of $q_1^{n_1/2} q_2^{n_2/2} q_3^{n_3/2}$ in the q -series $\mathbf{F}_a$ by $\mathbf{F}_a[q_1^{n_1/2} q_2^{n_2/2} q_3^{n_3/2}]$, this equation can be written as

$$\begin{aligned} \widehat{\mathbf{F}}_a(h_1, h_2, h_3, c)[q_1^{n_1/2} q_2^{n_2/2} q_3^{n_3/2}] \\ = \sum_{r_i + s_i = n_i} (-1)^{\sum_{i < j} (r_i s_j + s_i r_j)} \mathbf{F}_F[q_1^{r_1/2} q_2^{r_2/2} q_3^{r_3/2}] \mathbf{F}_a(h_1, h_2, h_3, c)[q_1^{s_1/2} q_2^{s_2/2} q_3^{s_3/2}] \end{aligned} \quad (7.21)$$

Using the decomposition of the representation $\mathcal{M}_{\text{NS}}$ under $\text{Vir} \oplus \text{Vir} \subset \text{SCA}_\nu \otimes \mathbf{F}$, we can also express $\widehat{\mathbf{F}}_a(h_1, h_2, h_3, c)$ in terms of Virasoro blocks. Using the fact that a basis of state of a $\text{SCA}_{1/2} \otimes \mathbf{F}$ module is given by $\{L_{-A}^{(1)} L_{-B}^{(2)} v_n\}$, one have

$$\begin{aligned} \widehat{\mathbf{F}}_a(h_1, h_2, h_3, c) = \sum_{\substack{|A|=|A'|, |B|=|B'|, |C|=|C'| \\ |D|=|D'|, |E|=|E'|, |F|=|F'| \\ n_1, n_2, n_3 \in \frac{1}{2}\mathbb{Z} \\ (-1)^{2(n_1+n_2+n_3)} = (-1)^a}} q_1^{2n_1^2+|A|+|B|} q_2^{2n_2^2+|C|+|D|} q_3^{2n_3^2+|E|+|F|} \eta_1^{2n_1} \eta_2^{2n_2} \eta_3^{2n_3} \\ \times (-1)^{4n_1 n_2 + 4n_1 n_3 + 4n_2 n_3} \frac{(G_{h_{1,n_1}}^{(1)})^{AA'} (G_{h_{1,n_1}}^{(2)})^{BB'} (G_{h_{2,n_2}}^{(1)})^{CC'} (G_{h_{2,n_2}}^{(2)})^{DD'} (G_{h_{3,n_3}}^{(1)})^{EE'} (G_{h_{3,n_3}}^{(2)})^{FF'}}{\|v_{1,n_1}\|^2 \|v_{2,n_2}\|^2 \|v_{3,n_3}\|^2} \\ \times \hat{\rho}_a(L_{-A}^{(1)} L_{-B}^{(2)} v_{1,n_1}, L_{-C}^{(1)} L_{-D}^{(2)} v_{2,n_2}, L_{-E}^{(1)} L_{-F}^{(2)} v_{3,n_3}) \hat{\rho}_a(L_{-A'}^{(1)} L_{-B'}^{(2)} v_{1,n_1}, L_{-C'}^{(1)} L_{-D'}^{(2)} v_{2,n_2}, L_{-E'}^{(1)} L_{-F'}^{(2)} v_{3,n_3}), \end{aligned} \quad (7.22)$$

where we have used the fact that $h_n^{(1)} + h_n^{(2)} - h = 2n^2$ and that v_n has Grassmann parity $2n$. G_h^{AB} is the Virasoro inverse Gram matrix. The resolution of $(-1)^{p_{\text{SCA}} \cdot p_{\mathbf{F}}}$ leads to the factorization of the Gram matrix.

The 3-point function $\hat{\rho}_a(L_{-A}^{(1)} L_{-B}^{(2)} v_{1,n_1}, L_{-C}^{(1)} L_{-D}^{(2)} v_{2,n_2}, L_{-E}^{(1)} L_{-F}^{(2)} v_{3,n_3})$ can be reduced to those of Virasoro primaries $\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})$ by conformal Ward identity. The two copies of Virasoro can be reduced independently, which leads to the factorization formula

$$\begin{aligned} \hat{\rho}_a(L_{-A}^{(1)} L_{-B}^{(2)} v_{1,n_1}, L_{-C}^{(1)} L_{-D}^{(2)} v_{2,n_2}, L_{-E}^{(1)} L_{-F}^{(2)} v_{3,n_3}) \\ = \hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3}) \rho(L_{-A} \nu_{h_{1,n_1}}^{(1)}, L_{-C} \nu_{h_{2,n_2}}^{(1)}, L_{-E} \nu_{h_{3,n_3}}^{(1)}) \rho(L_{-B} \nu_{h_{1,n_1}}^{(2)}, L_{-D} \nu_{h_{2,n_2}}^{(2)}, L_{-F} \nu_{h_{3,n_3}}^{(2)}), \end{aligned} \quad (7.23)$$

where $\rho(L_{-A} \nu_1, L_{-B} \nu_2, L_{-C} \nu_3)$ is defined in section 1. If we define the branching coefficient

$$B_a(h_1, h_2, h_3; n_1, n_2, n_3) = \frac{\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})}{\|v_{1,n_1}\| \|v_{2,n_2}\| \|v_{3,n_3}\|}, \quad (7.24)$$

the $\text{SCA}_\nu \otimes \mathbf{F}$ block can be decomposed into

$$\begin{aligned} \widehat{\mathbf{F}}_a(h_1, h_2, h_3, c) = \sum_{\substack{n_1, n_2, n_3 \in \frac{1}{2}\mathbb{Z} \\ (-1)^{2(n_1+n_2+n_3)} = (-1)^a}} q_1^{2n_1^2} q_2^{2n_2^2} q_3^{2n_3^2} \eta_1^{2n_1} \eta_2^{2n_2} \eta_3^{2n_3} (-1)^{4n_1 n_2 + 4n_1 n_3 + 4n_2 n_3} \\ \times [B_a(h_1, h_2, h_3; n_1, n_2, n_3)]^2 F(h_{1,n_1}^{(1)}, h_{2,n_2}^{(1)}, h_{3,n_3}^{(1)}, c^{(1)}) F(h_{1,n_1}^{(2)}, h_{2,n_2}^{(2)}, h_{3,n_3}^{(2)}, c^{(2)}), \end{aligned} \quad (7.25)$$

where $F(h_1, h_2, h_3, c)$ is the Virasoro conformal block, which can be computed efficiently using the c -relation relation in section 1. Since the fermion block $\mathbf{F}_F$ will always have the identity term 1, one can find $\mathbf{F}_F^{-1}$ as a formal power series in q , such that $\mathbf{F}_F \star \mathbf{F}_F^{-1} = 1$. The all-NS superconformal block $\mathbf{F}_a$ can then be computed as

$$\begin{aligned} \mathbf{F}_a(h_1, h_2, h_3, c) &= (\mathbf{F}_F)^{-1} \star \sum_{\substack{n_1, n_2, n_3 \in \frac{1}{2}\mathbb{Z} \\ (-1)^{2(n_1+n_2+n_3)} = (-1)^a}} q_1^{2n_1^2} q_2^{2n_2^2} q_3^{2n_3^2} \eta_1^{2n_1} \eta_2^{2n_2} \eta_3^{2n_3} (-1)^{4n_1n_2+4n_1n_3+4n_2n_3} \\ &\times [B_a(h_1, h_2, h_3; n_1, n_2, n_3)]^2 F(h_{1,n_1}^{(1)}, h_{2,n_2}^{(1)}, h_{3,n_3}^{(1)}, c^{(1)}) F(h_{1,n_1}^{(2)}, h_{2,n_2}^{(2)}, h_{3,n_3}^{(2)}, c^{(2)}). \end{aligned} \quad (7.26)$$

The NS branching coefficient

The remaining problem is to compute the branching coefficient $B_a(h_1, h_2, h_3; n_1, n_2, n_3)$. This has been worked out in [1312.4520, 1406.3008] (they called it the blow up factor). To compute this factor, let us define the normalization of v_n by the following.

We first introduce a free field representation of the $\text{SCA}_{1/2}$ algebra. The generators of the free field algebra $\mathcal{A}_{1/2}$ involve free boson operators $\{c_n | n \neq 0\}$, free fermion operators $\{\eta_r | r \in \mathbb{Z} + \frac{1}{2}\}$ (called ψ_r in [1608.02568]) and the central elements $\hat{P}$. They satisfy the algebraic relations

$$[c_n, c_m] = n\delta_{m+n,0}, \quad \{\eta_r, \eta_s\} = \delta_{r+s,0}. \quad (7.27)$$

We denote by $\mathcal{V}(P)$ the Fock representation of the free-field algebra on which the central element $\hat{P}$ acts as P . It is generated by acting c_{-n}, η_{-r} on the lowest weight vector $\phi(P)$, with $\hat{P}\phi(P) = P\phi(P)$.

The free field representation of $\text{SCA}_{1/2}$ is obtained by

$$\begin{aligned} L_n &= \frac{1}{2} \sum_{k \neq 0, n} c_k c_{n-k} + \frac{1}{2} \sum_r r \eta_{n-r} \eta_r + \frac{i}{2} (Qn - 2\hat{P}) c_n, \quad n \neq 0, \\ L_0 &= \sum_{k > 0} c_{-k} c_k + \sum_{r > 0} r \eta_{-r} \eta_r + \frac{1}{2} \left(\frac{Q^2}{4} - \hat{P}^2 \right), \\ G_r &= \sum_{n \neq 0} c_n \eta_{r-n} + i(Qr - \hat{P}) \eta_r. \end{aligned} \quad (7.28)$$

The Fock representation $\mathcal{V}(P)$ is therefore also a representation of $\text{SCA}_{1/2}$. For $P \notin \frac{1}{2}(b\mathbb{Z} + b^{-1}\mathbb{Z})$, this representation is irreducible as a $\text{SCA}_{1/2}$ representation and is isomorphic to the highest weight representation $\mathcal{V}_{\text{SCA}_{1/2}}(h)$ with $h = \frac{1}{8}Q^2 - \frac{1}{2}P^2$.

Consider the tensor-product representation $\mathcal{M}(P) := \mathcal{V}_{F_{1/2}} \otimes \mathcal{V}(P)$, where $\mathcal{V}_{F_{1/2}}$ is the Fock representation of the auxiliary free fermion. Since $\mathcal{M}(P)$ is a representation of $F_{1/2} \oplus \text{SCA}_{1/2}$

and two copy of virasoro algebra $\text{Vir} \oplus \text{Vir}$ is a subalgebra of $\mathbf{F}_{1/2} \oplus \text{SCA}_{1/2}$, it is a representation of $\text{Vir} \oplus \text{Vir}$.

For $P \notin \frac{1}{2}(b\mathbb{Z} + b^{-1}\mathbb{Z})$, $\mathcal{M}(P)$ is isomorphic to $\mathcal{M}_{NS}(h)$ with $h = \frac{1}{8}Q^2 - \frac{1}{2}P^2$. Under this identification, $\mathcal{M}(P)$ admits two natural bases. One is the Fock basis inherited from $\mathcal{M}(P)$, obtained by acting the free field generators $c_{-n}, \eta_{-r}, \psi_{-s}$ on the lowest weight vector $\mathbf{1}_F \otimes \phi(P)$. Another basis is the PBW basis inherited from $\mathcal{M}_{NS}(h)$, obtained by acting the super-virasoro generators $L_{-n}, G_{-r}, \psi_{-s}$ on $\mathbf{1}_F \otimes \phi(P)$. One can thus consider the base transformation between Fock basis and PBW basis, which a prior would be complicated.

For our application, we will always endow $\mathcal{M}(P)$ with the BPZ pairing, under which $L_n^\dagger = L_{-n}, G_r^\dagger = G_{-r}$. Under this pairing, one can speak of the BPZ conjugation of free field generators c_n, η_r , which generically does not admit a simple formula in terms of the original free field generators. Due to this reason, it is not straightforward to compute the norm of the state in the Fock basis. The efficient way to compute the norm is to do the base transformation to the PBW basis and then compute the norm there.

Yutai: I would actually expect that $P^\dagger = P$ under BPZ pairing.

The $\text{Vir} \oplus \text{Vir}$ primaries for $n > 0$ in $\mathcal{M}(P)$ admits a simple formula in the Fock basis

$$w_n(P) = \chi_{-\frac{4n-1}{2}}(P) \chi_{-\frac{4n-3}{2}}(P) \cdots \chi_{-\frac{1}{2}}(P) \phi(P), \quad \chi_r = \psi_r - i\eta_r. \quad (7.29)$$

For each free boson mode c_n and free fermion mode η , one can directly solve them in terms of G_{-r}, L_{-n} and $\hat{P}$. We define the Virasoro primary w_n in $\mathcal{M}_{NS}(h)$ to be the state after replacing c, η by L, G modes, $\phi(P)$ by ϕ , and $\hat{P}$ takes the positive root of $\frac{1}{4}Q^2 - 2h$. If h is greater than $Q^2/8$, we take $P = ip$ such that $p > 0$. The two states $w_n(P)$ and w_n will have the same norm.

Under the BPZ inner product where $L_n^\dagger = L_{-n}, G_r^\dagger = G_{-r}$, we adopt the following conjugation rules for $c, \eta, \hat{P}$:

$$c_n^\dagger = c_{-n}, \quad \eta_r^\dagger = \eta_{-r}, \quad \hat{P}^\dagger = -\hat{P} \quad (7.30)$$

and hence the BPZ conjugate of $\chi_r(P)$ is $-\chi_{-r}(-P) = -\psi_{-r} + i\eta_{-r}(-P)$. The remaining step of computing $\|w_n\|^2$ is then determining the commutator $\{\chi_r(-P), \chi_s(P)\}$. This is worked out in [1312.4520] by analyzing the pole structure of this quantity in P , and it turns out that

$$\|w_n(P)\|^2 = (-1)^{2n} 2^{2n} \frac{\ell(2P, 4n)}{\ell(Q + 2P, 4n)} \quad (7.31)$$

where the ℓ function is defined by the following. For positive integer m ,

$$\ell(x, m) = \begin{cases} \prod_{\substack{r,s \geq 0, r+s < m \\ r+s \equiv 0(2)}} (x + rb + sb^{-1}), & m > 0 \text{ even} \\ 2^{1/8} \prod_{\substack{r,s \geq 0, r+s < m \\ r+s \equiv 1(2)}} (x + rb + sb^{-1}), & m > 0 \text{ odd} \end{cases} \quad (7.32)$$

For zero or negative integer m ,

$$\ell(x, 0) = 1, \quad \ell(x, m) = \begin{cases} (-1)^{-m/2} \ell(Q - x, -m), & m < 0 \text{ even} \\ \ell(Q - x, -m), & m < 0 \text{ odd} \end{cases} \quad (7.33)$$

This formula has been checked against direct computation of the state w_n up to level $n = \frac{3}{2}$.

We normalize the Virasoro primary v_n with positive n to be

$$v_n = 2^{-2n} \ell(Q + 2P, 4n) w_n, \quad (7.34)$$

where, again, P is taken to be the positive for small h and purely imaginary with positive imaginary part for large h . v_0 is defined to be ϕ , and v_n with negative n is defined from the reflection formula

$$v_n(P) = v_{-n}(-P). \quad (7.35)$$

The norm of v_n is given by

$$\|v_n\|^2 = (-1)^{2n} 2^{-2n} \ell(2P, 4n) \ell(Q + 2P, 4n). \quad (7.36)$$

The 3-point function $\hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})$ is also computed in [1312.4520]. Normalizing

$$\hat{\rho}_0(\phi_1, \phi_2, \phi_3) = 1, \quad \hat{\rho}_1(\phi_1, G_{-1/2}\phi_2, \phi_3) = 1. \quad (7.37)$$

In our convention, their result reads

$$\begin{aligned} \hat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3}) &= (-1)^{n_1+n_2+n_3} i^a 2^{-(n_1+n_2+n_3)} \\ &\times \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right) \ell\left(\frac{Q}{2} - P_1 + P_2 + P_3, 2(-n_1 + n_2 + n_3)\right) \\ &\times \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right) \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right). \end{aligned} \quad (7.38)$$

For $(-1)^{2(n_1+n_2+n_3)} \neq (-1)^a$, the 3-point function is 0 by definition. Again, we should emphasize that this result only works for positive n_i . For negative n_i , one should use the reflection relation $v_n(P) = v_{-n}(-P)$ to convert it to a positive value. The values of P_i is the same in the definition of v_n . This formula has been checked directly against low-level ($n = 0, 1, \frac{1}{2}$) 3-point functions.

Putting everything together, we arrive at the final form for the NS branching coefficient:

$$\begin{aligned} B_a(h_1, h_2, h_3; n_1, n_2, n_3) &= \frac{(-i)^a}{\prod_{i=1}^3 \sqrt{\ell(2P_i, 4n_i) \ell(Q + 2P_i, 4n_i)}} \\ &\times \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right) \ell\left(\frac{Q}{2} - P_1 + P_2 + P_3, 2(-n_1 + n_2 + n_3)\right) \\ &\times \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right) \ell\left(\frac{Q}{2} + P_1 + P_2 + P_3, 2(n_1 + n_2 + n_3)\right). \end{aligned} \quad (7.39)$$

Checks against c -recursion

As a check of the formalism developed in this section, we compute the all-NS block $\mathbf{F}_a(h_1, h_2, h_3, c)$ as a q -expansion independently using the c -recursion relation and the double Virasoro decomposition. For all the vacuum blocks appearing in $\mathbf{F}, \mathbf{F}_F$ and F , we use the Schottky frame so that the Weyl anomaly factor on both sides agrees. We have tested numerically that, up to total level 4, the all-NS theta channel block $\mathbf{F}_a$ computed from the two methods agrees to machine precision for all 8 independent choices of $\{\eta_i\}$ and a .

8 Ramond blocks from double Virasoro

We now generalize the discussion above to Ramond blocks. Denoting the fermion ground states by u^a with $a = 0, 1$, normalizing

$$\langle u^0 | u^0 \rangle = 1, \quad \langle u^1 | u^1 \rangle = -1, \quad \psi_0 u^a = \frac{1}{\sqrt{2}} u^{1-a}, \quad (8.1)$$

the ramond $\text{SCA}_0 \otimes \mathbf{F}_0$ block is defined as

$$\begin{aligned} \widehat{\mathbb{F}}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c) = & \sum_{\substack{|A|=|A'|, |B|=|B'|, |C|=|C'| \\ (-1)^{A+B+C+|\alpha|+|\gamma|}=(-1)^f \\ (-1)^{A'+B'+C'+|\alpha'|+|\gamma'|}=(-1)^f \\ (-1)^{A+B+C+b+c}=1}} q_1^{|A|} \eta_1^A q_2^{|B|} \eta_2^{B+|\alpha|} q_3^{|C|} \eta_3^{C+|\gamma|} \mathbf{G}_{h_1}^{AA'} \mathbb{G}_{\beta_2}^{B\alpha, B'\alpha'} \mathbb{G}_{\beta_3}^{C\gamma, C'\gamma'} \\ & \times (-1)^{\text{sgn}(A)\text{sgn}(B)+\text{sgn}(A)\text{sgn}(C)+\text{sgn}(B)\text{sgn}(C)} q_1^{|A|} \eta_1^A q_2^{|B|} \eta_2^{B+b} q_3^{|C|} \eta_3^{C+c} \\ & \times \hat{\rho}_f^{(\eta)}(\Psi_{-A} \mathbf{L}_{-A} \phi_1, \Psi_{-B} u^b \otimes \mathbb{L}_{-B} w_2^\alpha, \Psi_{-C} u^c \otimes \mathbb{L}_{-C} w_3^\gamma) \\ & \times \hat{\rho}_f^{(\eta')}(\Psi_{-A} \mathbf{L}_{-A'} \phi_1, \Psi_{-B} u^b \otimes \mathbb{L}_{-B'} w_2^{\alpha'}, \Psi_{-C} u^c \otimes \mathbb{L}_{-C'} w_3^{\gamma'}) \end{aligned} \quad (8.2)$$

where Ψ_{-B} denotes a chain of R-sector fermion lowering operators (there are no $\mathbf{mathbb{b}}$ version of Ψ , sadly). Note that Ψ_{-A} does not contain the fermion zero mode ψ_0 . We have used $\langle u^a | \Psi_{-A}^\dagger \Psi_{-B} | u^b \rangle = (-1)^{A+a} \delta_{AB} \delta^{ab}$, as well as the convention $\langle u_1 \otimes x_1 | u_2 \otimes x_2 \rangle = \langle u_1 | u_2 \rangle \langle x_1 | x_2 \rangle$ without the additional operator exchanging sign. The shorthand sgn is defined such that $\text{sgn}(A) = A + \mathbf{A} \pmod{2}$ for NS sector and $\text{sgn}(B) = B + \mathbf{B} + \mathbf{b} + |\alpha| \pmod{2}$ for R sector. We have also used the fact that the fermion correlator is 0 unless $(-1)^{A+B+C+b+c} = 1$.

The 3-point function can be factorized as (we treat this formula as the definition of $\hat{\rho}_f^{(\eta)}$)

$$\begin{aligned} & \hat{\rho}_f^{(\eta)}(\Psi_{-A} \mathbf{L}_{-A} \phi_1, \Psi_{-B} u^b \otimes \mathbb{L}_{-B} w_2^\alpha, \Psi_{-C} u^c \otimes \mathbb{L}_{-C} w_3^\gamma) \\ & = (-1)^{A\mathbf{A}+(B+|\alpha|)(C+c)} \rho(\Psi_{-A} \mathbf{1}_F, \Psi_{-B} u^b, \Psi_{-C} u^c) \rho_f^{(\eta)}(\mathbf{L}_{-A} \phi_1, \mathbb{L}_{-B} w_2^\alpha, \mathbb{L}_{-C} w_3^\gamma) \end{aligned} \quad (8.3)$$

Therefore, if we define the fermion Ramond block as

$$\mathbb{F}_F = \sum_{A+B+C+b+c \equiv 0 \pmod{2}} q_1^{|A|} \eta_1^A q_2^{|B|} \eta_2^{B+b} q_3^{|C|} \eta_3^{C+c} (-1)^{A(B+b)+A(C+c)+(B+b)(C+c)} \times [\rho(\Psi_{-A} \mathbf{1}_F, \Psi_{-B} u^b, \Psi_{-C} u^c)]^2, \quad (8.4)$$

one then have the decomposition

$$\widehat{\mathbb{F}}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c) = \mathbb{F}_F \star_R \mathbb{F}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c)$$

where the R-sector convolution acts on monomials of q_i and η_i as

$$\begin{aligned} & q_1^{r_1} q_2^{r_2} q_3^{r_3} \eta_1^{\epsilon_1} \eta_2^{\epsilon_2} \eta_3^{\epsilon_3} \star_R q_1^{s_1} q_2^{s_2} q_3^{s_3} \eta_1^{\delta_1} \eta_2^{\delta_2} \eta_3^{\delta_3} \\ &= (-1)^{\sum_{i < j} (\epsilon_i \delta_j + \delta_i \epsilon_j)} q_1^{r_1+s_1} q_2^{r_2+s_2} q_3^{r_3+s_3} \eta_1^{\epsilon_1+\delta_1} \eta_2^{\epsilon_2+\delta_2} \eta_3^{\epsilon_3+\delta_3}. \end{aligned} \quad (8.5)$$

Using the decomposition (7.15), one can see that every state in $\mathcal{M}(\beta)$ can be written as $L_{-A}^{(1)} L_{-B}^{(2)} v_n^\alpha$, with $\alpha = \pm$ and $n \in \frac{1}{2}\mathbb{Z} + \frac{1}{4}$. The state v_n^α has conformal weight $h_n^{(1)} + h_n^{(2)} - h = 2n^2 - \frac{1}{16}$ and fermion parity $(-1)^\alpha$. Therefore,

$$\begin{aligned} \widehat{\mathbb{F}}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c) &= \sum_{\substack{|A|=|A'|, |B|=|B'|, |C|=|C'| \\ |D|=|D'|, |E|=|E'|, |F|=|F'| \\ n_1, n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4} \\ (-1)^{2n_1+\alpha_2+\alpha_3} = (-1)^f}} q_1^{2n_1^2+|A|+|B|} q_2^{2n_2^2-\frac{1}{8}+|C|+|D|} q_3^{2n_3^2-\frac{1}{8}+|E|+|F|} \eta_1^{2n_1} \eta_2^{\alpha_2} \eta_3^{\alpha_3} \\ &\times (-1)^{2n_1\alpha_2+2n_1\alpha_3+\alpha_2\alpha_3} \frac{(G_{h_{1,n_1}}^{(1)})^{AA'} (G_{h_{1,n_1}}^{(2)})^{BB'} (G_{h_{2,n_2}}^{(1)})^{CC'} (G_{h_{2,n_2}}^{(2)})^{DD'} (G_{h_{3,n_3}}^{(1)})^{EE'} (G_{h_{3,n_3}}^{(2)})^{FF'}}{\|v_{1,n_1}\|^2 \|v_{2,n_2}^{\alpha_2}\|^2 \|v_{3,n_3}^{\alpha_3}\|^2} \\ &\times \hat{\rho}_f^{(\eta)}(L_{-A}^{(1)} L_{-B}^{(2)} v_{1,n_1}, L_{-C}^{(1)} L_{-D}^{(2)} v_{2,n_2}^{\alpha_2}, L_{-E}^{(1)} L_{-F}^{(2)} v_{3,n_3}^{\alpha_3}) \hat{\rho}_f^{(\eta')}(L_{-A'}^{(1)} L_{-B'}^{(2)} v_{1,n_1}, L_{-C'}^{(1)} L_{-D'}^{(2)} v_{2,n_2}^{\alpha_2}, L_{-E'}^{(1)} L_{-F'}^{(2)} v_{3,n_3}^{\alpha_3}), \end{aligned} \quad (8.6)$$

where we used the fact that u^a has conformal weight $\frac{1}{16}$ and the fact that $\langle v_n^0 | v_n^1 \rangle = 0$. The factorization relation of the 3-point function reads

$$\begin{aligned} & \hat{\rho}_f^{(\eta)}(L_{-A}^{(1)} L_{-B}^{(2)} v_{1,n_1}, L_{-C}^{(1)} L_{-D}^{(2)} v_{2,n_2}^{\alpha_2}, L_{-E}^{(1)} L_{-F}^{(2)} v_{3,n_3}^{\alpha_3}) \\ &= \hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3}) \rho(L_{-A} \nu_{h_{1,n_1}}^{(1)}, L_{-C} \nu_{h_{2,n_2}}^{(1)}, L_{-E} \nu_{h_{3,n_3}}^{(1)}) \rho(L_{-B} \nu_{h_{1,n_1}}^{(2)}, L_{-D} \nu_{h_{2,n_2}}^{(2)}, L_{-F} \nu_{h_{3,n_3}}^{(2)}), \end{aligned} \quad (8.7)$$

and therefore,

$$\begin{aligned} \widehat{\mathbb{F}}_f^{(\eta, \eta')}(h_1, \beta_2, \beta_3, c) &= \sum_{\substack{n_1, n_2, n_3 \in \frac{1}{2}\mathbb{Z} + \frac{1}{4} \\ (-1)^{2n_1+\alpha_2+\alpha_3} = (-1)^f}} q_1^{2n_1^2} q_2^{2n_2^2-\frac{1}{8}} q_3^{2n_3^2-\frac{1}{8}} \eta_1^{2n_1} \eta_2^{2n_2} \eta_3^{2n_3} \\ &\times (-1)^{2n_1\alpha_2+2n_1\alpha_3+\alpha_2\alpha_3} \mathbb{B}_f^{(\eta)}(h_1, \beta_2, \beta_3; n_1, n_2, n_3) \mathbb{B}_f^{(\eta')}(h_1, \beta_2, \beta_3; n_1, n_2, n_3) \\ &\times F(h_{1,n_1}^{(1)}, h_{2,n_2}^{(1)}, h_{3,n_3}^{(1)}, c^{(1)}) F(h_{1,n_1}^{(2)}, h_{2,n_2}^{(2)}, h_{3,n_3}^{(2)}, c^{(2)}), \end{aligned} \quad (8.8)$$

where the R-sector branching coefficients are defined as

$$\mathbb{B}_f^{(\eta)}(h_1, \beta_2, \beta_3; n_1, n_2, n_3) = \frac{\hat{\rho}_f^{(\eta)}(v_{1,n_1}, v_{2,n_2}^{\alpha_2}, v_{3,n_3}^{\alpha_3})}{\|v_{1,n_1}\| \|v_{2,n_2}^{\alpha_2}\| \|v_{3,n_3}^{\alpha_3}\|}. \quad (8.9)$$

A Fusion polynomials and factorization formulae

We collect the result for all useful fusion polynomials and factorization formulae in this appendix. They are taken directly from the menu of a fusion restaurant in Cambridge, MA. All the formulas here are tested against direct low-level computations and checked to be consistent with the Mobius tranformation that permutes the three slots.

A.1 Fusion polynomials on a (NS, NS, NS) sphere

The All-NS fusion polynomials are defined as

$$\begin{aligned} \mathbf{P}_{rs}^0(h_i, h_j; c) &= \prod_{\substack{p=1-r, 1-r+2, \dots, r-1 \\ q=1-s, 1-s+2, \dots, s-1 \\ p+q-(r+s) \in 4\mathbb{Z}+2}} \frac{\lambda_i - \lambda_j + pb + qb^{-1}}{2\sqrt{2}} \frac{\lambda_i + \lambda_j + pb + qb^{-1}}{2\sqrt{2}}, \\ \mathbf{P}_{rs}^1(h_i, h_j; c) &= \prod_{\substack{p=1-r, 1-r+2, \dots, r-1 \\ q=1-s, 1-s+2, \dots, s-1 \\ p+q-(r+s) \in 4\mathbb{Z}}} \frac{\lambda_i - \lambda_j + pb + qb^{-1}}{2\sqrt{2}} \frac{\lambda_i + \lambda_j + pb + qb^{-1}}{2\sqrt{2}}. \end{aligned} \quad (\text{A.1})$$

They satisfy the reflection relation

$$\begin{aligned} \mathbf{P}_{rs}^0(h_i, h_j; c_{rs}) &= (-1)^{\lceil rs/2 \rceil} \mathbf{P}_{rs}^0(h_j, h_i; c_{rs}), \\ \mathbf{P}_{rs}^1(h_i, h_j; c_{rs}) &= (-1)^{\lfloor rs/2 \rfloor} \mathbf{P}_{rs}^1(h_j, h_i; c_{rs}). \end{aligned} \quad (\text{A.2})$$

For a null state in the first slot, the factorization relation is given by

$$\begin{aligned} &\rho_f(\mathbf{L}_{-A}\chi_{rs}, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3) \\ &= (-1)^{rsE} \mathbf{P}_{rs}^f(h_3, h_2; c_{rs}) \begin{cases} \rho_f(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3), & rs \in 2\mathbb{Z}, \\ \rho_{1-f}(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_3), & rs \in 2\mathbb{Z} + 1. \end{cases} \end{aligned} \quad (\text{A.3})$$

For the second slot,

$$\begin{aligned} &\rho_f(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\chi_{rs}, \mathbf{L}_{-E}\phi_3) \\ &= \mathbf{P}_{rs}^f(h_3, h_1; c_{rs}) \begin{cases} \rho_f(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-E}\phi_3), & rs \in 2\mathbb{Z}, \\ \rho_{1-f}(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-E}\phi_3), & rs \in 2\mathbb{Z} + 1, \end{cases} \end{aligned} \quad (\text{A.4})$$

and for the third slot,

$$\begin{aligned} &\rho_f(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\chi_{rs}) \\ &= (-1)^{rs(A+C)} \mathbf{P}_{rs}^f(h_1, h_2; c_{rs}) \begin{cases} \rho_f(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_{h_{rs}+rs/2}), & rs \in 2\mathbb{Z}, \\ \rho_{1-f}(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_{h_{rs}+rs/2}), & rs \in 2\mathbb{Z} + 1. \end{cases} \end{aligned} \quad (\text{A.5})$$

For the case where two of the three states are null states, one have

$$\begin{aligned}
\rho_f(\mathbf{L}_{-A}\chi_{rs}, \mathbf{L}_{-C}\chi_{rs}, \mathbf{L}_{-E}\phi_3) &= (-1)^{rsE} \mathbf{P}_{rs}^f(h_3, h_{rs}; c_{rs}) \mathbf{P}_{rs}^{f+rs} \left(h_3, h_{rs} + \frac{rs}{2}; c_{rs} \right) \\
&\quad \times \rho_f(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-E}\phi_3), \\
\rho_f(\mathbf{L}_{-A}\chi_{rs}, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\chi_{rs}) &= (-1)^{rs(f+rs)} \mathbf{P}_{rs}^f(h_{rs}, h_2; c_{rs}) \mathbf{P}_{rs}^{f+rs} \left(h_{rs} + \frac{rs}{2}, h_2; c_{rs} \right) \\
&\quad \times \rho_f(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-C}\phi_2, \mathbf{L}_{-E}\phi_{h_{rs}+rs/2}), \\
\rho_f(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\chi_{rs}, \mathbf{L}_{-E}\chi_{rs}) &= (-1)^{rs(A+C)} \mathbf{P}_{rs}^f(h_{rs}, h_1; c_{rs}) \mathbf{P}_{rs}^{f+rs} \left(h_1, h_{rs} + \frac{rs}{2}; c_{rs} \right) \\
&\quad \times \rho_f(\mathbf{L}_{-A}\phi_1, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbf{L}_{-E}\phi_{h_{rs}+rs/2}).
\end{aligned} \tag{A.6}$$

A.2 NS null states on a (NS, R, R) sphere

The two fusion polynomial for a NS null state on a (NS, R, R) sphere is defined as

$$\begin{aligned}
\mathbb{P}_{rs}^{\text{NS}, \eta}(\beta_2, \beta_3; c) &= \prod_{\substack{0 \leq k \leq r-1, 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}}} \frac{2\sqrt{2}(\beta_3 - \eta\beta_2) - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}} \\
&\quad \times \prod_{\substack{0 \leq k \leq r-1, 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}+1}} \frac{2\sqrt{2}(\beta_3 + \eta\beta_2) - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}}.
\end{aligned} \tag{A.7}$$

For the NS null state in the first slot, the factorization relation reads

$$\begin{aligned}
\rho_f^{(\eta)}(\mathbf{L}_{-A}\chi_{rs}, \mathbb{L}_{-C}w_{\beta_2}^\alpha, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \\
= \mathbb{P}_{rs}^{\text{NS}, \eta}(\beta_2, \beta_3; c_{rs}) \begin{cases} \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-C}w_{\beta_2}^\alpha, \mathbb{L}_{-E}w_{\beta_3}^\gamma), & rs \in 2\mathbb{Z}, \\ e^{i(-1)^f \pi/4} \rho_{1-f}^{(-\eta)}(\mathbf{L}_{-A}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-C}w_{\beta_2}^\alpha, \mathbb{L}_{-E}w_{\beta_3}^\gamma), & rs \in 2\mathbb{Z}+1. \end{cases}
\end{aligned} \tag{A.8}$$

For the third slot,

$$\begin{aligned}
\rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbb{L}_{-C}w_{\beta_2}^\gamma, \mathbf{L}_{-E}\chi_{rs}) \\
= \mathbb{P}_{rs}^{\text{NS}, (-1)^f \eta}(\beta_2, \beta_1; c_{rs}) \begin{cases} \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbb{L}_{-C}w_{\beta_2}^\gamma, \mathbf{L}_{-E}\phi_{h_{rs}+rs/2}), & rs \in 2\mathbb{Z}, \\ \eta e^{i(-1)^f \pi/4} \rho_{1-f}^{(\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbb{L}_{-C}w_{\beta_2}^\gamma, \mathbf{L}_{-E}\phi_{h_{rs}+rs/2}), & rs \in 2\mathbb{Z}+1. \end{cases}
\end{aligned} \tag{A.9}$$

Finally, for the second slot with even rs , one have

$$\begin{aligned}
\rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbf{L}_{-C}\chi_{rs}, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \\
= \frac{1}{2} \sum_{\epsilon=\pm 1} \mathbb{P}_{rs}^{\text{NS}, \epsilon}(\beta_3, \beta_1; c_{rs}) \left[\rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \right. \\
\left. - i\eta\epsilon \rho_f^{(-\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \right].
\end{aligned} \tag{A.10}$$

and for odd rs ,

$$\begin{aligned} & \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbf{L}_{-C}\chi_{rs}, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \\ &= \frac{(-1)^{(rs-1)/2} i^f e^{i\pi/4}}{2} \sum_{\epsilon=\pm 1} \mathbb{P}_{rs}^{\text{NS},\epsilon}(\beta_3, \beta_1; c_{rs}) \left[\eta \rho_{1-f}^{(\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \right. \\ & \quad \left. + i\epsilon \rho_{1-f}^{(-\eta)}(\mathbb{L}_{-A}w_{\beta_1}^\alpha, \mathbf{L}_{-C}\phi_{h_{rs}+rs/2}, \mathbb{L}_{-E}w_{\beta_3}^\gamma) \right]. \end{aligned} \quad (\text{A.11})$$

A.3 Ramond null state on a (NS, R, R) sphere

The two fusion polynomials associated with a Ramond null state are defined as

$$\begin{aligned} \mathbb{P}_{rs}^{\text{R},\eta}(h_i, \beta_j; c) &= \prod_{\substack{0 \leq k \leq r-1, \ 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}}} \frac{\lambda_i - 2\sqrt{2}\eta\beta_j - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}} \\ &\times \prod_{\substack{0 \leq k \leq r-1, \ 0 \leq \ell \leq s-1 \\ k+\ell \in 2\mathbb{Z}+1}} \frac{\lambda_i + 2\sqrt{2}\eta\beta_j - (1-r+2k)b - (1-s+2\ell)b^{-1}}{2\sqrt{2}}. \end{aligned} \quad (\text{A.12})$$

The factorization relations are

$$\begin{aligned} \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}\chi_{rs}^\alpha, \mathbb{L}_{-E}w_{\beta_j}^\gamma) &= \mathbb{P}_{rs}^{\text{R},\eta}(h_i, \beta_j; c_{rs}^{\text{R}}) \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}w_{\beta'_{rs}}^\alpha, \mathbb{L}_{-E}w_{\beta_j}^\gamma), \\ \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}w_{\beta_j}^\gamma, \mathbb{L}_{-E}\chi_{rs}^\alpha) &= \mathbb{P}_{rs}^{\text{R},\eta}(h_i, \beta_j; c_{rs}^{\text{R}}) \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}w_{\beta_j}^\gamma, \mathbb{L}_{-E}w_{\beta'_{rs}}^\alpha), \\ \rho_f^{(\eta)}(\mathbb{L}_{-A}\chi_{rs}^\alpha, \mathbb{L}_{-C}w_{\beta_j}^\gamma, \mathbf{L}_{-E}\phi_i) &= \mathbb{P}_{rs}^{\text{R},(-1)^f\eta}(h_i, \beta_j; c_{rs}^{\text{R}}) \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^\alpha, \mathbb{L}_{-C}w_{\beta_j}^\gamma, \mathbf{L}_{-E}\phi_i), \\ \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_j}^\gamma, \mathbb{L}_{-C}\chi_{rs}^\alpha, \mathbf{L}_{-E}\phi_i) &= \mathbb{P}_{rs}^{\text{R},(-1)^f\eta}(h_i, \beta_j; c_{rs}^{\text{R}}) \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_j}^\gamma, \mathbb{L}_{-C}w_{\beta'_{rs}}^\alpha, \mathbf{L}_{-E}\phi_i), \end{aligned} \quad (\text{A.13})$$

and for the (R, NS, R) order,

$$\begin{aligned} & \rho_f^{(\eta)}(\mathbb{L}_{-A}\chi_{rs}^\alpha, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta_j}^\gamma) \\ &= \frac{(-1)^{rs/2}}{2} \sum_{\epsilon=\pm 1} \mathbb{P}_{rs}^{\text{R},\epsilon}(h_i, \beta_j; c_{rs}^{\text{R}}) \left[\rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^\alpha, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta_j}^\gamma) \right. \\ & \quad \left. - i\eta\epsilon \rho_f^{(-\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^\alpha, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta_j}^\gamma) \right], \\ & \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_j}^\gamma, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}\chi_{rs}^\alpha) \\ &= \frac{(-1)^{rs/2}}{2} \sum_{\epsilon=\pm 1} \mathbb{P}_{rs}^{\text{R},\epsilon}(h_i, \beta_j; c_{rs}^{\text{R}}) \left[\rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta_j}^\gamma, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta'_{rs}}^\alpha) \right. \\ & \quad \left. - i\eta\epsilon \rho_f^{(-\eta)}(\mathbb{L}_{-A}w_{\beta_j}^\gamma, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta'_{rs}}^\alpha) \right]. \end{aligned} \quad (\text{A.14})$$

For the case where the two Ramond states are simultaneously null state,

$$\begin{aligned}
& \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}\chi_{rs}^\alpha, \mathbb{L}_{-E}\chi_{rs}^\gamma) \\
&= \mathbb{P}_{rs}^{\mathbf{R},\eta}(h_i, \beta_{rs}; c_{rs}^{\mathbf{R}}) \mathbb{P}_{rs}^{\mathbf{R},\eta}(h_i, \beta'_{rs}; c_{rs}^{\mathbf{R}}) \rho_f^{(\eta)}(\mathbf{L}_{-A}\phi_i, \mathbb{L}_{-C}w_{\beta'_{rs}}^\alpha, \mathbb{L}_{-E}w_{\beta'_{rs}}^\gamma), \\
& \rho_f^{(\eta)}(\mathbb{L}_{-A}\chi_{rs}^\alpha, \mathbb{L}_{-C}\chi_{rs}^\gamma, \mathbf{L}_{-E}\phi_i) \\
&= \mathbb{P}_{rs}^{\mathbf{R},(-1)^f\eta}(h_i, \beta_{rs}; c_{rs}^{\mathbf{R}}) \mathbb{P}_{rs}^{\mathbf{R},(-1)^f\eta}(h_i, \beta'_{rs}; c_{rs}^{\mathbf{R}}) \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^\alpha, \mathbb{L}_{-C}w_{\beta'_{rs}}^\gamma, \mathbf{L}_{-E}\phi_i), \\
& \rho_f^{(\eta)}(\mathbb{L}_{-A}\chi_{rs}^\alpha, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}\chi_{rs}^\gamma) \\
&= \frac{1}{2} \sum_{\epsilon=\pm 1} \mathbb{P}_{rs}^{\mathbf{R},\epsilon}(h_i, \beta_{rs}; c_{rs}^{\mathbf{R}}) \mathbb{P}_{rs}^{\mathbf{R},\epsilon}(h_i, \beta'_{rs}; c_{rs}^{\mathbf{R}}) \left[\begin{aligned} & \rho_f^{(\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^\alpha, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta'_{rs}}^\gamma) \\ & - i\eta\epsilon \rho_f^{(-\eta)}(\mathbb{L}_{-A}w_{\beta'_{rs}}^\alpha, \mathbf{L}_{-C}\phi_i, \mathbb{L}_{-E}w_{\beta'_{rs}}^\gamma) \end{aligned} \right].
\end{aligned} \tag{A.15}$$